Wang *et al. Int J Educ Technol High Educ* (2025) 22:59
https://doi.org/10.1186/s41239-025-00547-9

International Journal of Educational Technology in Higher Education

RESEARCH ARTICLE Open Access

In GenAI we trust: An investigation of university students' reliance on and resistance to generative AI in language learning

Feifei Wang[1], Nuoen Li[2], Alan C. K. Cheung[3*] and Gary K. W. Wong[1]

*Correspondence: alancheung@cuhk.edu.hk

[1] Centre for Information Technology in Education, Faculty of Education, The University of Hong Kong, Pokfulam, Hong Kong, China
[2] Department of Curriculum and Instruction, Faculty of Education, The Chinese University of Hong Kong, Shatin, Hong Kong, China
[3] Department of Educational Administration and Policy, Faculty of Education, The Chinese University of Hong Kong, Shatin, Hong Kong, China

Abstract

Advances in generative artificial intelligence (GenAI) have the potential to transform learning. As research increasingly calls for transforming the human-GenAI relationship from mere tool use to a collaborative partnership, there remains a notable gap in exploring an important factor influencing this relationship: trust in GenAI, being crucial amidst growing concerns about disuse, misuse, and abuse of GenAI in learner-GenAI interaction primarily via language. This study seeks to address the gap by investigating university students' trust in GenAI, its related factors of reliance and resistance, and boundary conditions in using GenAI for language learning. The research utilized an explanatory sequential mixed-methods design. In Study 1, a proposed conceptual model within the stimulus-organism-response framework was quantitatively tested through structural equation modeling using a survey. Data were collected from 682 university students who used GenAI in language learning. Study 2 further confirmed the results through 40 qualitative interviews and refined the conceptual model by offering a deeper understanding of the relationships. The findings indicate that trust in GenAI is a significant factor in university students' use of GenAI for language learning, with its influence on behavioral intention to use mediated through reliance and resistance. Moreover, forms of the five factors are identified: trust in GenAI, reliance, resistance, perceived risk, and actual use. Additionally, two boundary conditions are identified: perceived risk between trust in GenAI and reliance/resistance, and self-directed learning between behavioral intention to use and actual use. These findings provide theoretical, research, and practical implications regarding the appropriate use of GenAI in education.

Keywords: Generative artificial intelligence, Trust, Reliance, Resistance, Education

"No AI is an island" (Johnson & Vera, 2019). Artificial intelligence (AI) does not work in isolation from people; rather, generative AI (GenAI) is set to become integral to society (Lorenz et al., 2023), showing the potential to transform how human learning is delivered, cultivated, and evaluated (Yan et al., 2024). The integration of GenAI in learning has initiated a paradigmatic shift in human-AI relationships from human-AI stimulus–response to human-AI collaboration (Schelble et al., 2024). In human-AI collaboration for learning, GenAI's role is transitioning from merely a tool to a partner (Glikson &

Springer Open

© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

Woolley, 2020; Li et al., 2024; Yan et al., 2024). Learners are also expected to either infer knowledge in new contexts created by GenAI or co-create new knowledge with GenAI (Wang et al., 2024b; Yan et al., 2024). The new collaborative relationship requires learners to develop appropriate trust in GenAI to ensure its appropriate use throughout the learning process, given that trust is a central component in human-AI interaction (Choi et al., 2023; Choung et al., 2023; Jacovi et al., 2021). The importance of understanding learners' trust in GenAI should not be overstated in the current era of GenAI, especially when a growing body of research is highlighting students' disuse, misuse, and abuse of GenAI in education (e.g., Klingbeil et al., 2024; Meyer et al., 2023; Shahzad et al., 2024).

Scholars have indicated that students' disuse, misuse, and abuse of GenAI, which are associated with their inappropriate levels of trust and trust-related factors of reliance and resistance, may bring harmful consequences (Dwivedi et al., 2023; Jacovi et al., 2021). For one thing, when students disuse GenAI, showing distrust and resistance to this powerful technology, they are likely to miss out on its benefits, such as personalized support, diverse learning resources, and timely feedback in the widening digital divide (Yan et al., 2024). For another, when students use GenAI, with their trust in GenAI exceeding GenAI's trustworthiness, this may lead to misuse or abuse, resulting in a failure to monitor GenAI or biased decision-making (Jacovi et al., 2021). Common examples include the spread of misinformation in GenAI-generated content and the use of GenAI to cheat on exams by generating answers to questions (Dwivedi et al., 2023). When students misuse or abuse GenAI, they show over-reliance on it (Küper & Krämer, 2024). The over-reliance may lead students to overlook GenAI's flaws, thereby raising ethical concerns such as transparency, plagiarism, and privacy (Zhai et al., 2024). These harmful consequences impair effective learning, erode academic integrity, diminish achievement motivation, undermine cognitive abilities such as problem-solving and critical thinking, and damage the credibility of educational institutions (Abbas et al., 2024; Higgs & Stornaiuolo, 2024; Meyer et al., 2023).

Current research on trust in GenAI has been mostly documented in fields other than education, such as computer science (e.g., Schelble et al., 2024), public policy (e.g., Kreps et al., 2023), and management (e.g., Glikson & Woolley, 2020). While the research indicates that trust in GenAI influences individuals' reliance on or resistance to GenAI's responses (Klingbeil et al., 2024), existing research in the field of education is limited, primarily focusing on teachers' trust in AI in relation to their acceptance of AI in teaching (e.g., Choi et al., 2023; Nazaretsky et al., 2022). Despite the increasing use of GenAI among university students, who are crucial stakeholders in integrating GenAI into learning, very limited empirical research has focused on factors that drive their GenAI use (Strzelecki, 2023). Less is known about university students' trust in GenAI in learning. As more and more theoretical discussions, interviews, commentaries, and reviews are calling for research on students' trust in GenAI (Higgs & Stornaiuolo, 2024; Hınız, 2024; Kelly et al., 2023; Law, 2024; Zhai et al., 2024), few empirical studies have investigated university students' trust in GenAI in learning. Such an investigation of university students' trust in GenAI is crucial for relevant stakeholders such as educators, researchers, policymakers, and educational institutions to understand students' reliance and resistance that lead to disuse, misuse, and abuse of GenAI, shedding light on the development of effective strategies to integrate

GenAI into teaching, learning, and policymaking (Abbas et al., 2024; Meyer et al., 2023).

Therefore, in response to the important needs, the study contributes to existing literature from three aspects. First, to the best of our knowledge, this study is the first empirical study in education that aims to explore the relationship between university students' trust in GenAI and its related factors of reliance and resistance. We focus on the use of GenAI in language learning, as language learning is a major area significantly affected by large language models like ChatGPT (Hınız, 2024), and this area stands at an inflection point calling for further research to ensure ethical, valid, and reliable use of GenAI (Pack & Maloney, 2023). Investigation in this highly influenced area at an inflection point may provide insights into other areas of education. Second, this study makes theoretical contributions by constructing and consolidating a refined conceptual model of learners' trust in GenAI. This model is grounded in the stimulus-organism-response framework, including the stimulus factor (i.e., trust in GenAI), the organism factors (i.e., reliance and resistance), the response factors (i.e., behavioral intention to use, self-directed learning, and actual use), and boundary conditions (i.e., perceived risk and self-directed learning). Third, this study addresses the need to deepen understanding of psychological and cognitive factors underlying university students' behavioral intention to use and actual use of GenAI in learning (Kelly et al., 2023; Law, 2024).

Based on the aforementioned academic discourse and the identification of the three salient gaps in both theory and practice from the literature review, this study seeks to address the following three research questions (RQs).

RQ1: Based on the stimulus-organism-response (SOR) framework, to what extent does the stimulus factor (i.e., trust in GenAI) relate to the organism factors (i.e., reliance and resistance) among university students who use GenAI for language learning?

RQ2: To what extent do the organism factors in the SOR framework (i.e., reliance and resistance) relate to the response factors (i.e., behavioral intention to use, self-directed learning, and actual use) among university students who use GenAI for language learning?

RQ3: What additional attributes describe the stimulus, organism, and response factors of the SOR framework, and the circumscribed relationships defined within RQ1 and RQ2?

Literature review

Use of GenAI in language learning

Studies on the use of GenAI in language learning can mainly be classified into two categories regarding productivity and psychological aspects (Law, 2024). The first category relates to productivity gains that have been produced by GenAI. Some studies have employed descriptive statistics and experimental methods to examine the impact of productivity gains from GenAI use on text generation, grammatical error

correction, and sentence and paragraph refinement (e.g., Guo et al., 2023; Zhang et al., 2024b). This category mainly indicated that GenAI was able to facilitate content generation, grading, and language enhancements, thereby improving productivity in language learning.

Studies in the second category have explored factors influencing the use of GenAI in language learning from the perspectives of learners. These studies indicated that multiple factors related to learners affected GenAI use. For instance, learners' behavioral intention to use and motivation concurrently predicted their actual use of GenAI (Zheng et al., 2024). Positive attitudes toward the usefulness of GenAI predicted actual use of GenAI (Liu & Ma, 2023). However, these studies have insufficiently explored psychological or cognitive factors that may influence learners' use of GenAI (Kelly et al., 2023).

Continuing in the second category, only a few studies have explored the noteworthy psychological factors of learners, i.e., trust in, reliance on, and resistance to GenAI, that affect language learners' use of GenAI, along with relevant cognitive factors. Trust in AI is particularly important for emerging technologies like GenAI in light of widespread concerns about data security and accuracy (Ma, 2024). Trust in GenAI is associated with reliance and resistance, which in turn influence their acceptance and actual use (Chavoshi & Hamidi, 2019). The limited existing studies have highlighted concerns about blind trust, over-reliance, and complete resistance to GenAI, in relation to the negative impact on cognitive abilities such as critical thinking, problem-solving, and creativity (e.g., Higgs & Stornaiuolo, 2024; Law, 2024; Santiago Jr et al., 2023). While the limited existing studies have enhanced our understanding of the potential influence of the three important psychological factors: trust in, reliance on, and resistance to GenAI, much less attention has been paid to how the three factors interact with each other.

Trust in GenAI in language learning

The nature of trust in AI

The concept of trust is crucial across many domains involving interaction between humans and other parties, whether the parties are other individuals, technologies, or automated systems. Much of the earlier research focused on the importance of interpersonal trust in facilitating communication and cooperation among individuals (e.g., Driscoll, 1978; Shapiro, 1987). According to Mayer et al. (1995), interpersonal trust typically involves evaluations of other individuals' abilities, benevolence (i.e., the extent to which the individual is believed to have positive intentions toward the trustor), and integrity (i.e., the extent to which the individual adheres to principles acceptable by the trustor). Next, with the advent of the Internet, research on trust expanded from a focus on interpersonal trust to encompass trust in technology (e.g., Benbasat & Wang, 2005; Venkatesh et al., 2012). Research on trust in technology emphasizes the importance of evaluating technology's functionality rather than abilities, helpfulness rather than benevolence, and reliability rather than integrity when determining the acceptance and use of technology (Mcknight et al., 2011). In a similar vein, research on trust in automated systems has extended further, driven by advancements in automated systems that were capable of actively selecting data, transforming information, and making decisions (Lee & See, 2004). The studies indicate that user trust in automated systems is a dynamic construct influenced by factors related to systems (e.g., reliability and transparency),

users, and the context in which the interaction occurs (e.g., Balfe et al., 2018; Merritt et al., 2013). Overall, the concept of trust has continuously evolved, with different foci from interpersonal trust to trust in technology, and subsequently to trust in automated systems.

The increasing cognitive complexity of technology explains a growing interest in trust in AI. Trust tends to be less salient in well-structured and stable environments, where the emphasis on order and stability reduces transactional uncertainty (Poppo et al., 2016). However, as technology has recently advanced to dynamic and self-adapting AI that can be applied in different learning contexts, learning environments become increasingly complex and uncertain (Wu et al., 2024). Therefore, trust in AI is critical in helping learners cope with the cognitive complexity and uncertainty in diverse learning environments (Choung et al., 2023; Lee & See, 2004).

Trust in AI is conceptualized as an individual's expectancy of AI's technically competent role performance in the interests of the individual, especially in tasks where AI's decision-making is important to the individual (Barber, 1983; Mayer et al., 1995; Solberg et al., 2022). The definition highlights the key elements of trust in AI, including (1) the belief that AI will help achieve the individual's goals, especially in important and uncertain contexts, (2) the individual's willingness to be vulnerable by delegating partial responsibility for actions to AI, and (3) the individual's acceptance of potential risks or consequences resulting from the delegation of responsibility (Choung et al., 2023; Glikson & Woolley, 2020; Lee & See, 2004).

The aforementioned definition distinguishes trust in AI from the related yet distinct concepts that may also influence use behavior of AI: awareness, familiarity, reliance/resistance, and confidence. First, awareness of AI is an individual's recognition or understanding of AI's existence, its potential applications in specific tasks, and the associated ethical and legal concerns (Scantamburlo et al., 2025). Individuals may have awareness of AI without necessarily trusting it, in which vulnerability is minimal or absent. Second, familiarity with AI refers to an individual's accumulated experience or knowledge gained through repeated interactions with AI (Arce-Urriza et al., 2025). Individuals who are familiar with AI do not necessarily trust AI, depending on the satisfaction of interactions with AI (Komiak & Benbasat, 2006). Third, reliance on or resistance to AI reflects an individual's positive or negative evaluations or attitudes, and willingness or reluctance to adopt AI (Klingbeil et al., 2024; van Dongen & van Maanen, 2013). Reliance on and resistance to AI, as attitudes, can be shaped by trust in AI as a belief (Cheng et al., 2022). Fourth, both confidence and trust in AI involve expectations that may lead to disappointment, but trust differs from confidence in that trust requires prior engagement and involves recognizing and accepting potential risks (Mayer et al., 1995). An individual can have confidence in AI without necessarily trusting it through prior engagement with AI. Differentiating trust in AI from the similar concepts further clarifies its nature.

The relationship between trust in AI and AI capability in defining the appropriate use of AI

Inappropriate trust in AI, which is associated with disuse, misuse, and abuse, partly depends on how well trust in AI aligns with AI capabilities (Lee & See, 2004). As Fig. 1 shows, supporting appropriate trust is important for the appropriate use of AI. Trust can be calibrated when an individual's trust in AI matches AI's objective capabilities.

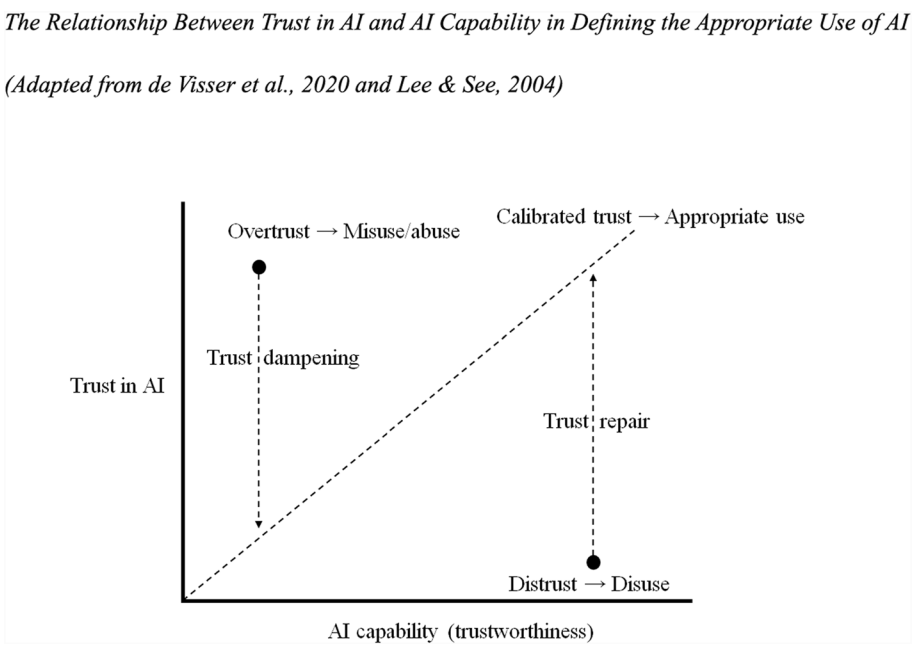

Fig. 1 The Relationship Between Trust in AI and AI Capability in Defining the Appropriate Use of AI (Adapted from de Visser et al., 2020 and Lee & See, 2004)

Overtrust represents poor calibration in which trust exceeds AI capabilities, and distrust reflects that trust falls short of AI capabilities. To calibrate appropriate trust related to the appropriate use of AI, when trust is too high, trust dampening strategies can be employed, such as demonstrating technology failures and providing likelihood alerts (de Visser et al., 2020). When trust is too low, trust repair strategies can be applied, such as clearly stating the level of user control and demonstrating AI proficiency (Israelsen & Ahmed, 2019). When trust in AI matches AI capabilities, AI can be used appropriately.

In educational settings, the appropriate use of AI occurs when individuals actively engage with AI, critically evaluate AI-generated outputs, maintain ethical awareness, transparently acknowledge the use of AI, and preserve their own agency during interactions with AI (Symeou et al., 2025). Conversely, the appropriateness of AI use can be compromised by disuse, misuse, and abuse (Jacovi et al., 2021). Disuse refers to avoidance of AI capabilities, even when its capabilities may support learning. For example, students who lack trust in GenAI may be less motivated to use it, even when supporting resources are available, as they focus on its limitations and hazards (Zhang et al., 2024a). Misuse refers to unintended or inappropriate use due to misunderstanding, a lack of relevant expertise, or insufficient critical judgment, potentially resulting in decision biases (Parasuraman & Riley, 1997). For example, learners might use ChatGPT in writing without verifying accuracy, unintentionally reinforcing incorrect writing usage (Dergaa et al., 2023). Abuse involves intentionally exploiting AI capabilities for personal benefits or unfair advantages, typically resulting in ethical violations (Kendall & Teixeira da Silva, 2023). For example, language learners might deliberately submit GenAI-generated essays as their original work, consciously committing plagiarism (Meyer et al., 2023). Disuse, misuse, and abuse of AI, all associated with trust in AI, have become important concerns impacting the appropriate use of AI in education.

Theoretical framework

The overarching theoretical framework is the stimulus-organism-response (SOR) framework, as extant studies have demonstrated its predictive power regarding university students' reactions to emerging technologies (e.g., Bai, 2025; Zhao et al., 2020). Additionally, we adopt theoretical constructs from Trust Calibration Theory (Lee & See, 2004; Parasuraman & Riley, 1997) and the Theory of Reasoned Action (Ajzen & Fishbein, 1980; Fishbein & Ajzen, 1975) to establish the relationships among constructs. The SOR framework suggests that various aspects of triggers (i.e., the stimulus) evoke intermediary psychological or cognitive states within individuals (i.e., the organism), which in turn influence their behavioral responses (i.e., the response) to the triggers (Mehrabian & Russell, 1974).

First, the stimulus component refers to factors that affect individuals' internal states (Mehrabian & Russell, 1974). The traditional SOR framework primarily emphasizes external stimuli influencing individuals, but recent research indicates that stimuli can also be internal, such as brand trust (Gong et al., 2023) and health anxiety (Goyal et al., 2021), particularly in contexts involving interaction with AI technology (Vafaei-Zadeh et al., 2024). In GenAI-assisted language learning, trust in GenAI impacts learners' confidence and subsequent decision to use the technology, while a lack of trust leads to skepticism and reduces the likelihood of use (Shahzad et al., 2024). Thus, trust in GenAI acts as a stimulus that influences learners' internal experiences.

Second, the organism component, including intermediary psychological or cognitive states, mediates the relationship between the stimulus and the response (Mehrabian & Russell, 1974). This study considers reliance on and resistance to GenAI as organism factors, based on Trust Calibration Theory that suggests that trust influences reliance on and resistance to automation (Lee & See, 2004; Parasuraman & Riley, 1997). The relationships have also been theoretically and empirically supported in other studies (e.g., Cheng et al., 2022; Klingbeil et al., 2024; Mayer et al., 1995; Solberg et al., 2022).

Third, the response component refers to the outcome, the reaction, or the final action (Mehrabian & Russell, 1974). The study includes behavioral intention to use, actual use, and self-directed learning as response factors. Empirical evidence indicates that in the final phase of the SOR framework, self-directed learning, defined as learners' ability to actively manage or guide their own learning behaviors, influences the relationship between behavioral intention to use and actual use (Esiyok et al., 2024; Lai et al., 2022).

Finally, the proposed factors in the SOR components have been supported by the Theory of Reasoned Action (Ajzen & Fishbein, 1980; Fishbein & Ajzen, 1975), which suggests that behaviors (i.e., actual use in this study) are driven by intentions (i.e., behavioral intention to use), and intentions are influenced by attitudes (i.e., reliance and resistance). Attitudes in turn are based on beliefs (i.e., trust in GenAI).

Conceptual model development and hypotheses for Study 1

Stimulus: trust in GenAI

As Fig. 2 shows, trust in AI is a major predictor that affects behavioral intention to use AI (Choi et al., 2023). Particularly with GenAI, trust is considered crucial because it inherently involves uncertainty and risks (Law, 2024). This characteristic of GenAI critically affects

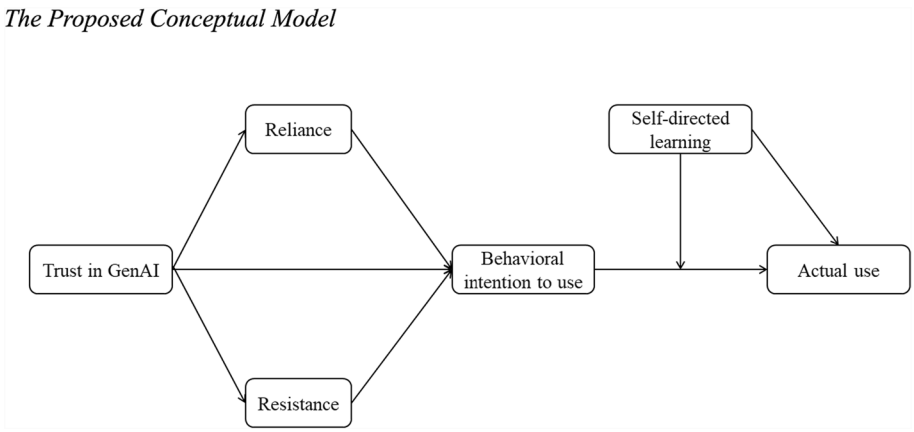

Fig. 2 The Proposed Conceptual Model

learners' trust, which may result in a lower acceptance rate (Higgs & Stornaiuolo, 2024). Possible reasons include the imperfect algorithms of GenAI, ethical dilemmas (e.g., transparency, equality, and beneficence), and disruptive assessment (Yan et al., 2024). Trust in GenAI may mitigate the adverse impact of uncertainty and risks, potentially influencing behavioral intention to use GenAI.

Trust in GenAI influences whether individuals rely on GenAI advice or resist it (Klingbeil et al., 2024). The emergence of GenAI and its increasing accessibility have shifted the dynamics of human-AI interaction, gradually transforming machines from mere tools to partners (C. Wang et al., 2024a, 2024c), and highlighting the importance of trust in GenAI. When used appropriately, GenAI has considerable potential to improve learning, so fostering appropriate trust is vital (Klingbeil et al., 2024). Excessive trust might lead to over-reliance, causing learners to blindly rely on GenAI, even when such trust is not warranted (Li & Little, 2023). Conversely, a lack of trust might result in resistance, leading to low rates of adoption or complete rejection of beneficial technology (Orbaiz & Arce-Urriza, 2024). Therefore, we hypothesize:

H1

Trust in GenAI positively influences behavioral intention to use.

H2

Trust in GenAI positively influences reliance.

H3

Trust in GenAI negatively influences resistance.

Organism: reliance on and resistance to GenAI

Reliance on or resistance to GenAI reflects individuals' positive or negative evaluations toward GenAI (Klingbeil et al., 2024; van Dongen & van Maanen, 2013). When individuals opt for GenAI judgments, they tend to show reliance on GenAI, and when they prefer their own judgments, they tend to display resistance to GenAI. Individuals' evaluations when choosing between GenAI and their own judgments are influenced by factors such as overconfidence, the quality of GenAI outputs, and the specific domains of judgments. Typically, individuals favor GenAI in domains that require concrete accuracy standards, and they prefer their own judgments in domains based on personal experience (Mahmud et al., 2022).

Regarding reliance on GenAI, it has been found that reliance positively influences learners' adoption of GenAI. The more reliance learners have on AI, the more likely they are to use it in learning (Seo et al., 2021). Furthermore, reliance on GenAI is a crucial intermediary between trust in GenAI and learners' adoption of GenAI (Chavoshi & Hamidi, 2019). Students with higher levels of trust in GenAI are more likely to rely on it, increasing their behavioral intention to use and actual use in practical learning activities (Shahzad et al., 2024). As for resistance to GenAI, it has been suggested that resistance negatively affects learners' use of GenAI, and potentially influences the link between trust in GenAI and adoption of GenAI (Higgs & Stornaiuolo, 2024). Learners with lower levels of trust in GenAI are more likely to resist it, decreasing their behavioral intention to use and actual use (Nazaretsky et al., 2022). Therefore, we hypothesize:

H4

Reliance positively influences behavioral intention to use.

H5

Resistance negatively influences behavioral intention to use.

H6

Trust in GenAI positively influences behavioral intention to use via reliance and resistance.

H7

Trust in GenAI positively influences actual use via behavioral intention to use, via reliance and behavioral intention to use, and via resistance and behavioral intention to use.

Response: behavioral intention to use, self-directed learning, and actual use

Behavioral intention to use refers to an individual's subjective probability of using a particular technology (Venkatesh et al., 2012). In GenAI-assisted language learning, behavioral intention to use means the degree to which learners intend to use GenAI in language learning. It is a significant predictor of actual use (Liu & Ma, 2023), which in

this study refers to students' practical implementation of GenAI in language learning. Language learners' actual practices of using GenAI are based on their behavioral intention to use GenAI for language learning purposes (Zheng et al., 2024).

Self-directed learning has been reconceptualized to include three aspects: learners' personal attributes (e.g., moral, emotional, and intellectual management), autonomous process (i.e., including learner autonomy over the learning process), and learning context (i.e., the environmental factors) that influence how learners use GenAI to learn language (Li et al., 2024; Song & Hill, 2007). Students with higher levels of self-directed learning have more positive personal attributes (e.g., attitudes and self-efficacy) and cognitive strategies, affecting their engagement and actual use of GenAI in learning (Li et al., 2024). Furthermore, self-directed learning can potentially influence the relationship between behavioral intention to use and actual use (Lai et al., 2022). Students with higher levels of self-directed learning are more likely to convert their behavioral intention to use into actual use. Therefore, we hypothesize:

H8

Behavioral intention to use positively influences actual use.

H9

Self-directed learning positively influences actual use.

H10

The relationship between behavioral intention to use and actual use is moderated by self-directed learning.

Figure 2 displays the conceptual model based on the developed hypotheses.

Method

This study employed a mixed-methods research design grounded in pragmatic philosophy, which is most appropriate given the nature of the RQs. Such a research design is not confined by a single research philosophy or paradigmatic assumptions, but instead offers inclusive, pluralistic, and complementary approaches that provide the best chance of obtaining useful answers by combining deduction (i.e., testing theories and hypotheses), induction (i.e., discovering patterns and rich meanings), and abduction (i.e., uncovering the best set of explanations for understandings results) within a pragmatic rationale (Johnson & Onwuegbuzie, 2004; Venkatesh et al., 2016). Specifically, this study followed an explanatory sequential mixed-methods design (Creswell & Plano Clark, 2017), where a quantitative phase was initially performed, followed by a subsequent qualitative phase to understand and extend the quantitative results in depth. This design is preferred in AI-related educational studies for identifying mechanisms that may remain obscured by purely quantitative approaches and for an in-depth exploration of findings from quantitative results through qualitative data analyses (Liu et al., 2024). RQ1 and RQ2 were

addressed through a survey in Study 1, and the results were further explored through a subsequent explanatory qualitative analysis in Study 2. Building upon the survey results, Study 2 addressed RQ3 with semi-structured interviews and a thematic analysis, qualitatively exploring the relationships posited within the theoretical framework (Fraenkel et al., 2018). The procedure for this explanatory sequential mixed-methods design is displayed in Table 1.

Participants

Study 1 adopted a purposive sampling strategy to recruit participants with an online survey. This sampling strategy ensures that participants are most likely to provide appropriate and useful information, maximizing the efficient use of limited research resources (Campbell et al., 2020). This study recruited eligible participants based on two criteria. First, they were students from Chinese universities. Second, they had previously used GenAI in language learning. Between June 2024 and August 2024, university students who self-identified as being from Chinese universities and had used GenAI in language learning were invited to participate. An anonymous link from Wenjuanxing, i.e., a widely used online survey platform in China, was distributed via social media tools such as WeChat, QQ, and Weibo. Of the 700 survey responses initially collected, 682 were valid and included in analyses after applying the two criteria. There were no missing data in the responses, as the submission was set on the online platform to be possible only after all items had been completed. Demographic information of the participants is shown in Table 2.

Table 1 The Procedure for the Explanatory Sequential Mixed-Methods Design *(Adapted from* Ivankova et al., 2006*)*

Research question (RQ)	Phase	Procedure	Product
RQ1 and RQ2 (Quantitative)	Quantitative data collection	Cross-sectional online survey ($N=682$) Purposive sampling	Numeric data
	Quantitative data analysis	Preliminary analyses Structural equation modeling AMOS 29.0 and IBM SPSS Statistics 26.0	Descriptive statistics Measurement model and structural model Mediation analysis, moderation analysis, and path analysis
	Bridging quantitative and qualitative phases	Developing interview questions	Interview guide
RQ3 (Qualitative)	Qualitative data collection	Individual semi-structured interviews ($N=40$) Purposive sampling: Quota sampling and maximal variation sampling	Text data: Interview transcripts and memos Audio recordings
	Qualitative data analysis	Thematic analysis Within-case and across-case comparisons of themes NVivo 14	Keywords, codes, and themes Similar and different themes Cross-thematic matrix
	Integrating quantitative and qualitative results	Interpreting and explaining quantitative and qualitative results	The refined conceptual model Theoretical, research, and practical implications

Table 2 *Demographic Information of Participants (Study 1: N = 682; Study 2: N = 40)*

Characteristic	Category	Frequency (Study 1)	Percentage (Study 1)	Frequency (Study 2)	Percentage (Study 2)
Gender	Male	297	43.55%	19	47.50%
	Female	385	56.45%	21	52.50%
Age	Below 16	4	0.59%	1	2.50%
	17–20	336	49.27%	21	52.50%
	21–24	285	41.79%	15	37.50%
	25–28	50	7.33%	2	5.00%
	Above 29	7	1.03%	1	2.50%
Education level	Undergraduate	587	86.07%	29	72.50%
	Postgraduate	95	13.93%	11	27.50%
Field of study	Arts & humanities	192	28.15%	15	37.50%
	Social sciences	153	22.43%	12	30.00%
	Science & engineering	256	37.54%	8	20.00%
	Medicine	81	11.88%	5	12.50%
Location of the university	Eastern China	130	19.06%	12	30.00%
	Southern China	207	30.35%	11	27.50%
	Western China	184	26.98%	9	22.50%
	Northern China	161	23.61%	8	20.00%
Average days per week using GenAI for language learning	Rarely	173	25.37%	7	17.50%
	1–2 days	245	35.92%	11	27.50%
	3–4 days	140	20.53%	9	22.50%
	5–6 days	55	8.06%	7	17.50%
	Every day	69	10.12%	6	15.00%
GenAI tool most used	Ernie Bot	173	25.37%	11	27.50%
	iFLYTEK	245	35.92%	9	22.50%
	ChatGPT	140	20.53%	11	27.50%
	Bard/Gemini	55	8.06%	6	15.00%
	Others	69	10.12%	3	7.50%

For Study 2, to develop an in-depth exploration of the results from Study 1, we adopted two purposive sampling strategies to select eligible participants from those in Study 1 who agreed to participate in Study 2 (Creswell & Guetterman, 2019). First, we employed quota sampling to ensure representation of key demographic categories (e.g., gender, age, and education level). Specifically, we aligned the quotas with the distributions in Study 1. Within each category, we calculated participants' summed mean scores and category means from survey responses, and then selected participants from each category whose mean scores fell within one standard error of the mean. This ensured that we selected participants who closely represented the category characteristics in each subgroup. Second, we used maximal variation sampling, in which we added participants with diverse characteristics within the categories. This approach allowed us to identify various dimensions of the categories and preserve multiple perspectives within the same categories. Altogether, as Table 2 displays, the two strategies generated a sample of 40 eligible participants, thereby trying to preserve multiple viewpoints while maintaining the demographic proportions from Study 1.

Table 3 Survey Items for the Factors

Factor	Item	Reference
Trust in GenAI	1. I feel that my decision-making process with GenAI is reliable in language learning.	Liu et al. (2015)
	2. I feel that my strategies used to reach decisions assisted with GenAI are trustworthy in language learning.	
	3. I feel that the information security of GenAI is creditable in language learning.	
	4. I feel that the efficacy to manage GenAI is believable in language learning.	
	5. Overall, I think that GenAI is trustworthy to me in language learning.	
Reliance	6. I depend on GenAI for decision-making in language learning.	Cheng et al. (2022)
	7. I depend on GenAI for new information or resources as my language learning input.	
	8. I depend on GenAI for feedback or evaluation on my language learning output.	
	9. I depend on GenAI for suggestions on my language learning process.	
Resistance	10. I oppose the change to the new way of learning language with GenAI.	Cheng et al. (2022); Kim & Kankanhalli (2009)
	11. I will try to avoid using GenAI in language learning.	
	12. I do not agree with the suggestions provided by GenAI in language learning.	
	13. I prefer not to use GenAI in my language learning.	
Behavioral intention to use	14. I expect to use GenAI frequently in the future for language learning.	Liu & Ma (2024); Sun & Mei (2022)
	15. I plan to use GenAI for my language learning needs in the future.	
	16. I would like to use GenAI to support my language learning in the future.	
	17. I think I will utilize GenAI for my language learning.	
Self-directed learning	18. If there is something I don't understand in a class, I always find a way to learn it on my own.	Lai et al. (2022); Lounsbury et al. (2009)
	19. I am good at finding the right resources to make learning more efficient and effective.	
	20. If there is something I need to learn, I find a way to do so right away.	
	21. I am very motivated to learn on my own without having to rely on other people.	

Table 3 (continued)

Factor	Item	Reference
Actual use	22. I use GenAI to help understand language learning materials.	Lai et al. (2022)
	23. I use GenAI to acquire more language knowledge.	
	24. I use GenAI to help express my thoughts.	
	25. I use GenAI to seek language learning strategies and tips.	
	26. I use GenAI to check my understanding.	
	27. I use GenAI to check my learning progress.	
	28. I use GenAI to expand opportunities to use language.	
	29. I use GenAI to sustain motivation and interest in language learning.	
	30. I use GenAI to ask for support and help.	

Quantitative data collection and analysis

Survey instruments

To collect quantitative data, this study adapted established questionnaires to measure all latent factors with 30 items to fit the specific context of the research, as detailed in Table 3. The survey was translated following a forward and backward translation procedure. In the forward translation, three bilingual translators whose native language was Chinese independently translated the survey from English to Chinese, and then discussed any discrepancies to reach a consensus. Next, two native English-speaking translators independently performed backward translations of the consensus Chinese version and provided comments for further modifications. Finally, all translators and the research team discussed and resolved any remaining discrepancies. The six-point Likert scale from "strongly disagree" to "strongly agree" was used to measure participants' responses. The even-numbered six-point scale was used to minimize risks related to deviations in responses, aiming to improve discrimination and reliability in responses from self-reported surveys (Chomeya, 2010). Pilot testing of the survey was conducted with 53 students from Chinese universities who had experience using GenAI in language learning. According to their feedback, a few modifications to the wording were made to enhance the clarity and comprehensiveness of the items.

Quantitative data analysis

Study 1 is based on relationship testing, so a quantitative analysis was conducted to test hypotheses. We analyzed the data following a two-step structural equation modeling (SEM) approach with AMOS 29.0 and IBM SPSS Statistics 26.0 (Collier, 2020; Kline, 2023). First, basic descriptive statistics of all measured factors were examined as preliminary analyses. Specifically, we assessed the normality of the data by

checking skewness and kurtosis of items. Following the initial analysis, we proceeded to assess reliability of the survey by computing Cronbach's alpha values. We then evaluated convergent validity of the factors through the standardized factor loading of each item, and the composite reliability (CR) and the average variance extracted (AVE) for each factor. We also examined discriminant validity between the factors by comparing inter-factor correlation coefficients with the square roots of AVE for the factors. Second, we used SEM with maximum likelihood parameter estimates to test the mediation effects of reliance and resistance in the relationship between university students' trust in GenAI and behavioral intentions to use, and the moderation effect of self-directed learning in the relationship between behavioral intentions to use and actual use. The mediation effects were achieved by estimating the measurement model and path analysis results (Collier, 2020). The indices used to measure the overall fit of the model included the chi-square statistic ($\chi 2$), the ratio of chi-square to degrees of freedom ($\chi 2/df$), the comparative fit index (CFI), the incremental fit index (IFI), the Tucker-Lewis index (TLI), the normative fit index (NFI), the standardized root mean square residual (SRMR), and the root mean square error of approximation (RMSEA) (Schreiber et al., 2006). According to the literature, the threshold levels for model fit indices that indicate adequacy are as follows: $\chi 2/df < 3$, CFI, IFI, TLI, and NFI > 0.9, and SRMR and RMSEA < 0.08 (Hair et al., 2010; Kline, 2023). For mediation analyses, we used bootstrapping with 5000 bootstrap resamples to assess indirect effects within mediation models, utilizing a two-tailed test, 95% bias-corrected confidence interval (CI), p-values, and user-defined estimands to determine the paths of indirect effects. For moderation analyses, we adopted the matched-pairs interaction approach to test moderation in a full structural model (Collier, 2020). This approach reduces the issues associated with the full indicator interaction and is thus recommended for evaluating moderation involving latent factors (Collier, 2020; Marsh et al., 2004).

Qualitative data collection and analysis

Study 2 aims to build upon and corroborate the quantitative results of Study 1, while providing deeper insights that enhance the theoretical and empirical undertones gained from Study 1. The final results from both studies are integrated into the refined conceptual model, as presented in Fig. 5.

Study design

Forty semi-structured interviews, lasting between 30 and 40 min, were conducted online to uncover the rich narrative meanings underlying the model proposed in Study 1. We first reviewed the participants' responses to the survey, and then conducted in-depth interviews using an interview guide (see Table 4) developed from the literature (Glikson & Woolley, 2020; Jacovi et al., 2021). Communication with the participants was designed to gain detailed insights into their trust in, reliance on, and resistance to GenAI, and use of GenAI for language learning. Such intensive communications are important for comprehending and consolidating theoretical frameworks (Xu & Zammit, 2020).

Table 4 Semi-Structured Interview Guide

Main questions	Examples of follow-up questions
1. Can you describe your initial thoughts or attitudes about using GenAI in language learning?	Are there changes in your initial thoughts or attitudes? Why or why not?
2. Do you trust GenAI in language learning? Why or why not?	What helps you decide whether GenAI's responses are trustworthy or untrustworthy in language learning? Does the trustworthiness or untrustworthiness influence how much you want to use GenAI in language learning?
3. Do you rely on GenAI in language learning?	In what ways do you find yourself relying on GenAI in language learning? How, if at all, does your trust in GenAI (or lack of trust) factor into that decision? What are other factors that influence that decision? When you feel reliance on using GenAI, how do you choose to use it for language learning?
4. Have there been times when you have been reluctant or refused to use GenAI for language learning? Why?	Are there specific reasons that lead you to trust GenAI less and potentially resist it in language learning? When you feel resistance to using GenAI, what alternatives do you choose to assist you in language learning? Are the alternatives working? Why or why not?
5. Do you prefer to take the initiative in learning if you want to learn something new or encounter difficulties in learning?	Does this initiative have any influence on your use of GenAI in language learning? Can you give an example?
6. Have you noticed any changes in the way you learn language since you started using GenAI?	If yes, what caused the changes? If no, what has prevented any noticeable change in the way you learn language?

Qualitative data analysis

A thematic analysis was performed to analyze the interviews (Creswell & Guetterman, 2019). Each interview was audio-taped, transcribed, and analyzed with NVivo 14. The analysis adheres to the guidelines proposed by Naeem et al. (2023) for conducting a thematic analysis to develop a conceptual model, including six steps: (1) transcribing, familiarizing with data, and selecting quotations, (2) selecting keywords, (3) coding, (4) constructing themes, (5) conceptualizing through interpreting keywords, codes, and themes, and (6) developing a conceptual model. The credibility of the results was established through separate coding by the authors, code verification by two independent researchers, and collaborative discussions among all the authors and independent researchers to resolve inconsistencies and reach a consensus (Roberts et al., 2019; Xu & Zammit, 2020).

Results

The quantitative results

Measurement model

Table 5 presents the descriptive results of 30 items. The item means ranged from 2.890 to 4.776, with the standard deviation values fluctuating between 0.838 and 1.478. The skewness and kurtosis indices for the items did not exceed |3.0| and |8.0|, respectively (Kline, 2023), which indicated that the normal distribution of the data was not violated. Therefore, SEM analyses were further performed.

The measurement model assessed the reliability and validity of the survey, ensuring that the dataset met the requisite assumptions for conducting advanced statistical tests.

Table 5 Descriptive Statistics and Details of Measurement Items

Factor and item	Mean	*SD*	Skewness	Kurtosis	Factor loading	Cronbach's alpha
Trust in GenAI						
Item 1	4.570	0.913	− 0.628	1.013	0.767	0.877
Item 2	4.510	0.892	− 0.280	0.167	0.783	
Item 3	4.468	0.882	− 0.262	0.070	0.726	
Item 4	4.588	0.891	− 0.342	0.269	0.782	
Item 5	4.579	0.921	− 0.381	0.363	0.772	
Reliance						
Item 6	4.196	1.069	− 0.260	− 0.246	0.735	0.869
Item 7	4.314	1.079	− 0.360	− 0.059	0.769	
Item 8	4.290	1.043	− 0.462	0.043	0.823	
Item 9	4.260	1.131	− 0.406	− 0.141	0.839	
Resistance						
Item 10	2.890	1.379	0.448	− 0.702	0.915	0.944
Item 11	2.899	1.412	0.381	− 0.683	0.886	
Item 12	2.916	1.367	0.321	− 0.872	0.898	
Item 13	3.016	1.478	0.378	− 0.793	0.897	
Behavioral intention to use						
Item 14	4.642	0.869	− 0.193	0.005	0.741	0.849
Item 15	4.689	0.867	− 0.438	0.344	0.772	
Item 16	4.711	0.838	− 0.211	− 0.231	0.770	
Item 17	4.776	0.866	− 0.258	− 0.423	0.778	
Self-directed learning						
Item 18	4.485	0.923	− 0.232	− 0.264	0.764	0.832
Item 19	4.585	0.930	− 0.436	0.130	0.791	
Item 20	4.570	0.893	− 0.262	0.048	0.712	
Item 21	4.604	0.908	− 0.235	− 0.249	0.708	
Actual use						
Item 22	4.738	1.251	− 1.344	1.669	0.878	0.963
Item 23	4.609	1.264	− 1.070	0.933	0.877	
Item 24	4.613	1.255	− 1.163	1.056	0.849	
Item 25	4.598	1.312	− 1.037	0.588	0.865	
Item 26	4.632	1.261	− 1.150	1.105	0.856	
Item 27	4.425	1.303	− 0.812	0.180	0.862	
Item 28	4.638	1.279	− 1.049	0.674	0.873	
Item 29	4.428	1.294	− 0.841	0.279	0.847	
Item 30	4.717	1.234	− 1.209	1.344	0.858	

SD standard deviation

Table 6 Convergent Validity and Discriminant Validity

	Factor	CR (> 0.7)	AVE (> 0.5)	1	2	3	4	5	6
1	Self-directed learning	0.832	0.554	**0.744**					
2	Actual use	0.963	0.745	0.173	**0.863**				
3	Behavioral intention to use	0.850	0.586	0.396	0.184	**0.766**			
4	Resistance	0.944	0.808	− 0.014	0.100	− 0.153	**0.899**		
5	Reliance	0.871	0.628	0.417	0.175	0.462	0.065	**0.792**	
6	Trust in GenAI	0.877	0.587	0.540	0.260	0.564	− 0.114	0.656	**0.766**

The square roots of AVEs are highlighted in bold. *CR* composite reliability. *AVE* average variance extracted

As shown in Table 5, the Cronbach's alpha values for the six factors ranged from 0.832 to 0.963, which were all higher than the threshold of 0.7 (Hair et al., 2010), indicating adequate internal reliability. To assess validity, we first computed the standardized factor loading of each item and the CR and AVE values of each factor to examine convergent validity. The standardized factor loadings were between 0.712 and 0.915, all higher than the threshold of 0.5 (Hair et al., 2010). As displayed in Table 6, the CR values were between 0.832 and 0.963, all higher than the 0.7 limit (Kline, 2023). The AVE values were between 0.554 and 0.808, all higher than the threshold of 0.5 (Kline, 2023). These results indicated the establishment of convergent validity. As for discriminant validity, Table 6 shows that the square root of the AVE for each factor was larger than the inter-factor correlation coefficients (Hair et al., 2010), thereby suggesting good discriminant validity.

The fitness of the model was evaluated through computing and checking the model-fit indices, including $\chi 2/df$, CFI, IFI, TLI, NFI, SRMR, and RMSEA. It was found that all the seven indices showed adequate values for the model as $\chi 2/df$ (2.435) < 3, CFI (0.953) > 0.9, IFI (0.953) > 0.9, TLI (0.949) > 0.9, NFI (0.923) > 0.9, SRMR (0.051) < 0.08, and RMSEA (0.046) < 0.08. Therefore, this model has a good fit.

Structural model

The structural model involves testing the structural relationships of the proposed hypotheses. In the path analysis, standardized regression weights were calculated, revealing the directions and relationships between factors. Relationships among the six factors were examined to test the hypotheses.

Table 7 and Fig. 3 present the results of SEM to examine the hypotheses. It was found that all the ten hypotheses were accepted. First, regarding direct effects, trust in GenAI positively influenced behavioral intention to use (H1: $\beta = 0.442$, $p < 0.001$)

Table 7 Hypothesis Testing

Hypothesis	Beta value	*SE*	Sig. value	Hypothesis result
Direct effect				
H1: TGAI → BIU	0.442	0.054	0.000	Accepted
H2: TGAI → RL	0.662	0.053	0.000	Accepted
H3: TGAI → RS	− 0.095	0.079	0.024	Accepted
H4: RL → BIU	0.176	0.046	0.002	Accepted
H5: RS → BIU	− 0.114	0.018	0.002	Accepted
H8: BIU → AU	0.149	0.076	0.000	Accepted
H9: SD → AU	0.123	0.070	0.006	Accepted
Indirect effect				
H6: TGAI → RL → BIU	0.117	0.044	0.009	Accepted
TGAI → RS → BIU	0.011	0.006	0.029	Accepted
H7: TGAI → BIU → AU	0.066	0.026	0.004	Accepted
TGAI → RL → BIU → AU	0.017	0.009	0.007	Accepted
TGAI → RS → BIU → AU	0.002	0.001	0.020	Accepted
Moderating effect				
H10: SD × BIU → AU	0.118	0.071	0.007	Accepted

TGAI trust in GenAI, *RL* reliance, *RS* resistance, *BIU* behavioral intention to use, *SD* self-directed learning, *AU* actual use, *SE* standard error, *Sig. value* significance value

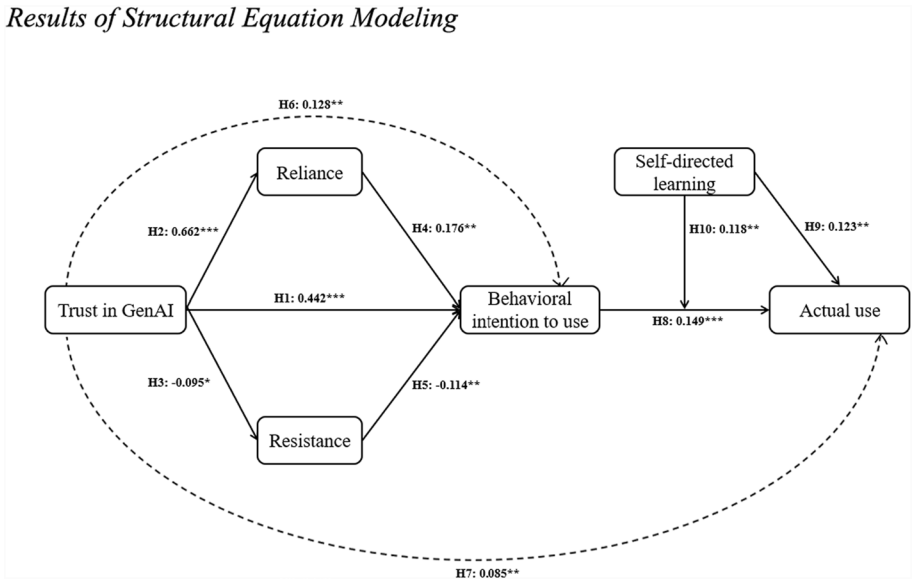

Fig. 3 Results of Structural Equation Modeling

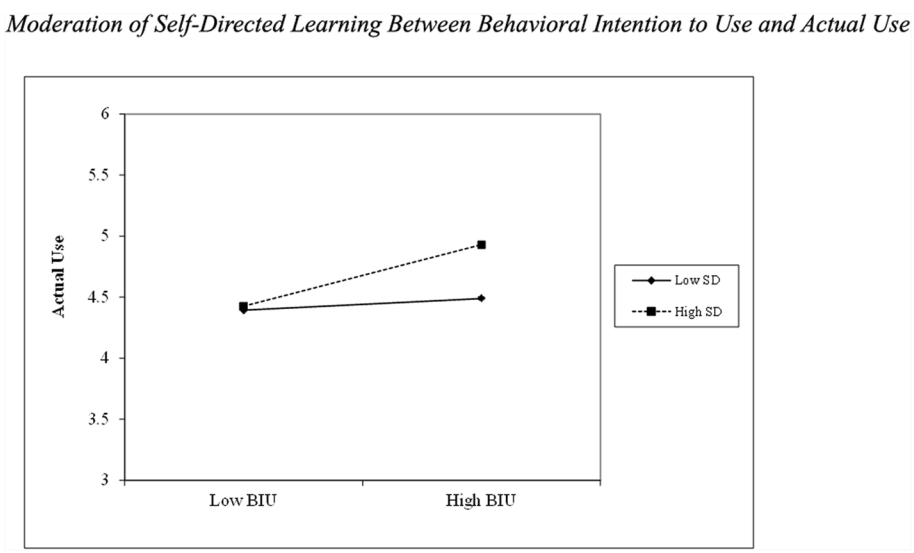

Fig. 4 Moderation of Self-Directed Learning Between Behavioral Intention to Use and Actual Use. *SD* self-directed learning. *BIU* behavioral intention to use

and reliance (H2: $\beta = 0.662$, $p < 0.001$). Trust in GenAI negatively influenced resistance (H3: $\beta = -0.095$, $p = 0.024$). Reliance positively influenced behavioral intention to use (H4: $\beta = 0.176$, $p = 0.002$). Resistance negatively influenced behavioral intention to use (H5: $\beta = -0.114$, $p = 0.002$). Behavioral intention to use positively influenced actual use (H8: $\beta = 0.149$, $p < 0.001$). Self-directed learning positively influenced actual use (H9: $\beta = 0.123$, $p = 0.006$). Second, as for indirect effects, trust in GenAI produced significant positive effects on behavioral intention to use through reliance (H6: $\beta = 0.117$, $p = 0.009$) and resistance (H6: $\beta = 0.011$, $p = 0.029$). Moreover, trust in GenAI produced significant positive effects on actual use via behavioral intention to use (H7: $\beta = 0.066$, $p = 0.004$),

via reliance and behavioral intention to use (H7: $\beta = 0.017$, $p = 0.007$), and via resistance and behavioral intention to use (H7: $\beta = 0.002$, $p = 0.020$). Third, concerning the moderating effect, the interaction term (self-directed learning × behavioral intention to use) had a significant effect on actual use (H10: $\beta = 0.118$, $p = 0.007$). The results of moderation were further explored through moderation graphs. Figure 4 also showed that there was significant moderation by self-directed learning between behavioral intention to use and actual use, as indicated by the non-parallel lines on the graph. This suggests that the impact of behavioral intention to use on actual use increases in the presence of self-directed learning.

The qualitative results

The qualitative data reinforced and extended the quantitative results of Study 1 by offering explanatory insights through the SOR framework. First, the qualitative analysis identified forms of trust in GenAI, reliance, resistance, perceived risk, and actual use. Second, the analysis provided explanations for students' reliance and resistance in relation to their trust in GenAI. Third, the qualitative findings explored the nuanced role of perceived risk in the relationship between trust in GenAI and reliance/resistance, and the role of self-directed learning in the relationship between behavioral intention to use and actual use. The refined conceptual model is shown in Fig. 5.

Stimulus: contextualizing trust in GenAI in relation to influences from GenAI, learners, and external factors

According to the interview transcripts, trust in GenAI has been contextualized in relation to influences from GenAI, learners, and external factors. A dynamic interplay among these three aspects shaped learners' trust in GenAI, influencing both intrinsic

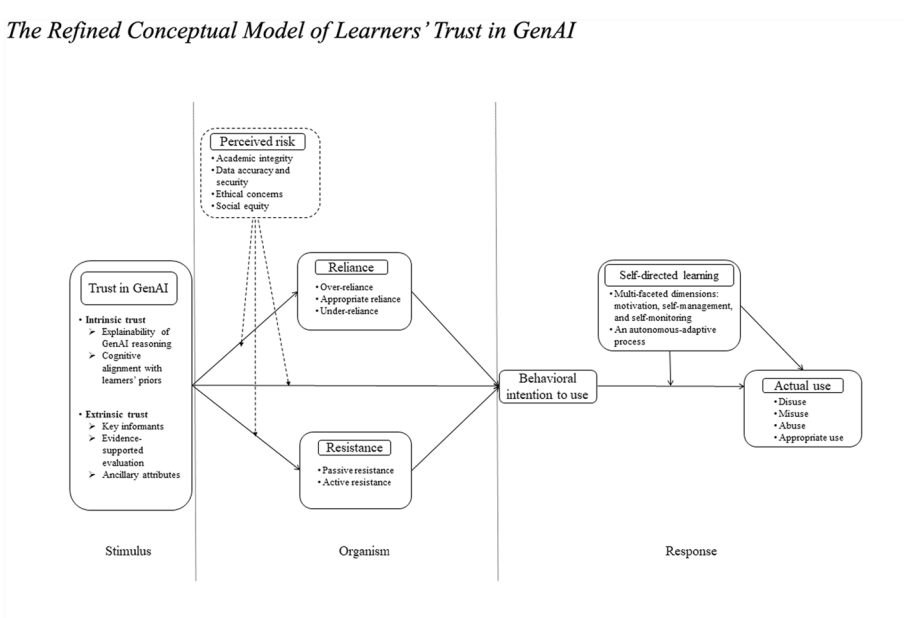

Fig. 5 The Refined Conceptual Model of Learners' Trust in GenAI

and extrinsic forms of trust. Intrinsic trust arose when the observable decision process of the GenAI model matched users' priors on how this process should operate. Extrinsic trust fundamentally relied on the evaluation scheme, specifically evaluating the GenAI model's ability to generalize in new scenarios based on its expected behavior in similar scenarios (Jacovi et al., 2021). In other words, intrinsic trust was trust in the decision process of the GenAI model, and extrinsic trust was trust in the evaluation methodology or data for the GenAI model.

Intrinsic trust emerged through the interaction between learners and GenAI, consisting of two codes: explainability of GenAI reasoning and cognitive alignment with learners' priors. The former code, explainability of GenAI reasoning, included four subcodes: comprehensive explanatory support, real-time interaction, consistent outputs, and transparent error handling. Comprehensive explanatory support was first stated by participants as important in building trust with GenAI in language learning. For instance, Student 27 (S27) stated, "When GenAI provides accessible and clear built-in explanations for something confusing in language learning, it builds my trust. It feels like having a reliable tutor by my side, helping me through complex parts of the language." Furthermore, real-time interaction and consistent outputs were emphasized. For instance, S40 stated, "When I receive immediate responses to my queries from GenAI, I feel that I can ask for help any time…The consistent outputs seem to reassure me of its dependability, since I can predict how it will respond in different scenarios." Interestingly, the last subcode of intrinsic trust, transparent error handling, also helped to increase trust, as stated by 72.5% of participants. S11 stated, "I know that GenAI is imperfect, so when it clearly explains why an error occurred, it not only helps me to understand what went wrong but also teaches me how to avoid similar errors in the future."

The latter code of intrinsic trust, cognitive alignment with learners' priors, involved four subcodes: familiarity with AI technology, cognitive load management, personalized feedback, and contextual appropriateness. Specifically, familiarity with AI technology, whether through understanding knowledge about AI or previous usage experience, enhanced the students' trust in GenAI for language learning. S28 stated, "I do not know much about the mechanism of GenAI, but I use it often and have tried out some useful tips along the way. This experience has made me trust it as a helpful tool for my studies." The second subcode of cognitive load management meant that the display of GenAI outputs was expected to align with students' cognitive abilities. S11 explained, "When GenAI presents information in a way that is neither too simple nor too complex, it helps me to absorb the information better without feeling underwhelmed or overwhelmed." Personalized feedback, as the third subcode, was stated by participants as crucial in building their trust in GenAI because personalization implied that GenAI understood their learning styles (S1, S18, and S24), learning goals (S16 and S22), proficiency levels (S3 and S31), pace of learning (S37), and motivations in language learning (S25 and S33). Finally, the last subcode, contextual appropriateness, was also emphasized when GenAI showed cultural sensitivity according to students' cultural norms and values (S30), specific learning tasks (S10), and ethical considerations such as promoting positive values (S4 and S31).

Another form, extrinsic trust, arose from the influence of external factors and GenAI itself. Among the 40 participants interviewed, nine students (i.e., 22.5%) reported

perceiving explicit support from their teachers or institutions for using GenAI in language learning, whereas 15 students (i.e., 37.5%) reported receiving no explicit support. The remaining 16 students (i.e., 40%) expressed uncertainty about their teachers' or institutions' stance toward GenAI use in language learning. Therefore, these results indicated mixed perceptions among participants regarding whether GenAI use for language learning was formally permitted by teachers or institutions. In this context, the participants' extrinsic trust included three codes: key informants, evidence-supported evaluation, and ancillary attributes. First, the opinions of key informants on GenAI reasoning, such as teachers, student peers, and policymakers, facilitated students' extrinsic trust in GenAI. For example, S22 explained, "When teachers share how effectively GenAI can be integrated into learning and demonstrate its benefits, it reassures me of its value in learning." Second, evidence-supported evaluation from various media sources was highlighted. As stated by S4, "When I read studies or news articles showing how GenAI has enhanced other students' language skills, this developed my trust in it as beneficial and well-tested." Third, ancillary attributes of GenAI played a role in trust-building, such as a user-friendly interface (S28 and S32), perceived emotional support from GenAI in learning (S12, S20, and S34), and the inclusion of anthropomorphic elements like human-like appearance (S6 and S17).

Organism: reliance and resistance based on trust in GenAI

Based on trust in GenAI, participants showed related reliance and resistance in language learning. On the one hand, reliance involved three forms: over-reliance, appropriate reliance, and under-reliance (Schemmer et al., 2023). On the decision-making continuum from self-reliance to AI reliance in learning, over-reliance occurred when students accepted GenAI feedback without critical evaluation, leading to misuse or abuse of GenAI (Jacovi et al., 2021). For example, S1 stated, "I accept GenAI's suggestions and corrections without taking the time to understand or evaluate them because I think its language proficiency is much higher than mine." Alternatively, appropriate reliance occurred when students critically evaluated GenAI feedback and effectively utilized it in learning, ensuring its appropriate use. S37 stated, "Instead of passively accepting GenAI feedback, I usually evaluate whether the feedback addresses my learning questions. I often revise my questions several times to obtain the answers I seek." Finally, under-reliance occurred when students primarily depended on themselves for decision-making, frequently overlooking GenAI feedback, even when it was accurate and beneficial, leading to misuse or abuse of GenAI. S33 stated, "When using GenAI, I check for its usefulness, but it often appears to fall short. However, I may miss some valuable insights, as there were times when I realized parts of its feedback were correct." To summarize, three forms of reliance were reflected: over-reliance, under-reliance, and appropriate reliance, leading to misuse, abuse, or the appropriate use.

On the other hand, resistance included two forms: passive and active resistance. Resistance, as a generic and negative evaluation of adopting innovations, is characterized by a reluctance to make changes associated with such adoption, and/or by satisfaction with the status quo. The potential for this rejection is either temporary or long-term (Orbaiz & Arce-Urriza, 2024). Among participants, passive resistance was mainly expressed in the behaviors of persistent adherence to former learning habits, coupled with potentially

temporary disuse of GenAI. Active resistance involved more pronounced behaviors such as expressing negative opinions and discouraging others from using, alongside potentially long-term disuse of GenAI in language learning. For instance, while explaining passive resistance, S29 stated, "I stick to traditional learning habits like using textbooks and handwritten notes because they make me feel I am really learning knowledge and encourage my deep thinking, whereas using GenAI appears to provide only superficial information." When talking about active resistance, S7 emphasized, "I am skeptical about using GenAI in language learning, often telling my friends that it is ineffective and misleading. Based on what I have seen, I will not use it in the future." Common reasons for resistance, either passive or active, included satisfaction with existing learning habits (S15 and S26), complexity of GenAI use (S8 and S23), and negative personal and/or observed use experiences (S14, S21, and S29).

Regarding the relationships between trust in GenAI and reliance/resistance, our interview transcripts revealed an association between participants' levels of trust in GenAI and their corresponding levels of reliance/resistance. Specifically, those who expressed high levels of trust often relied on GenAI without critical evaluation, sometimes to the point of over-reliance (e.g., S1 and S5). By contrast, those who expressed low levels of trust, often due to skepticism about GenAI's accuracy or ethical considerations, tended to actively resist using GenAI (e.g., S13 and S36). Moreover, those who expressed calibrated trust adopted a more balanced approach, appropriately relying on GenAI for specific tasks, such as grammar checks, while maintaining their own critical evaluation (e.g., S2 and S37).

However, the association between participants' levels of trust in GenAI and their corresponding levels of reliance/resistance was not always consistent, as a critical moderating factor emerged to shape these dynamics: perceived risk. The exploratory study also identified the boundary conditions of perceived risk that either mitigated or strengthened the influence of trust in GenAI on reliance, resistance, and behavioral intention to use. The perceived risk that the students highlighted was about academic integrity (e.g., plagiarism, education fairness, and evaluation appropriateness), data accuracy and security (e.g., data free from misinformation and protected against leaks), ethical concerns (e.g., perpetuation of biases against certain groups and irresponsible use of GenAI-generated content), and social equity (e.g., the digital divide and accessibility barriers). The perceived risk either weakened or enhanced the impact of trust in GenAI on reliance, resistance, and behavioral intention to use, demonstrating its vital role as a moderating factor.

Specifically, perceived risk was the boundary condition between trust in GenAI and the three factors: reliance, resistance, and behavioral intention to use. In relation to reliance, S11 stated, "When I have concerns about data security, my trust diminishes, making me hesitant to depend on GenAI. However, knowing that my data are well-protected restores my trust, enhancing my reliance." As for resistance, S35 explained, "I worry about plagiarism for the difficulty teachers face in distinguishing between student-created and GenAI-generated content, which decreases my trust and increases my resistance. However, plagiarism detection technologies at our university boost my trust and help reduce my resistance." In association with behavioral intention to use, S9 expressed, "The initial GenAI tool I used focused mainly on grammar and vocabulary corrections,

reducing my trust due to its shallow evaluations. Switching to another GenAI tool that could offer in-depth essay assessments increased my trust, making me more inclined to use it." These showed the important role of perceived risk in shaping students' interactions with GenAI, influencing how trust in GenAI affected reliance, resistance, and behavioral intention to use.

Responses: the moderating role of self-directed learning and forms of actual use

According to the SOR framework, stimulus (i.e., trust in GenAI) and organism factors (i.e., reliance and resistance) influence the subsequent response factors (i.e., behavioral intention to use, self-directed learning, and actual use). Similar to the stimulus-organism linkage, a boundary condition, self-directed learning, existed between behavioral intention to use and actual use. This meant that behavioral intention to use influenced actual use, but this influence was shaped by the students' self-directed learning. In the process, self-directed learning showed multi-faceted dimensions: motivation, self-management, and self-monitoring. As for motivation and self-management, S32 described, "GenAI allows me to customize my learning path and progress. No more feeling stuck or helpless. I can shape my learning just the way I need it, taking control and moving forward in my own attempts." As to self-monitoring, S38 expressed, "My interaction with GenAI is not merely about getting things right. It is also about ensuring everything ties together meaningfully. I feel like I am responsible for my learning journey." Overall, students with self-directed learning were motivated to take responsibility for constructing meaning with GenAI in self-monitoring and understanding, when they had some control over their learning experiences in self-management.

Self-directed learning was the boundary condition between behavioral intention to use and actual use, emerging as an autonomous and adaptive learning process. At one end, students exercised agency and took charge of their learning process while using GenAI, and at the other end, they adapted dynamically to GenAI feedback with repeated usage. When stating the moderating role of self-directed learning between behavioral intention to use and actual use, S39 stated, "I value self-management. When improving my vocabulary, I use GenAI to practice and track progress. This approach transforms my attempts into daily habits, bridging the gap between planning and doing." These highlighted the importance of self-directed learning for influencing the transition from behavioral intention to use to actual use.

Regarding the students' actual use of GenAI in language learning, the interview transcripts suggested that they used GenAI for a range of language learning tasks. Specifically, the participants primarily reported using GenAI to assist with academic writing (e.g., course assignments), professional writing (e.g., emails), creative writing (e.g., stories and poems), brainstorming new ideas, translating texts, summarizing lengthy passages, proofreading, and language learning strategies. For instance, S18 remarked, "I often use ChatGPT to summarize long passages, which helps me to quickly grasp the key ideas and save time when reviewing course materials." Several other participants noted that they frequently used GenAI for initial proofreading to identify spelling and structural errors before conducting more thorough edits themselves (e.g., S6, S24, and S38). These findings highlight GenAI's broad utility in facilitating various aspects of the

language learning process, from pre-writing ideation to the refinement of final written outputs.

Within the broad range of GenAI utilities in language learning, four distinct forms of actual use of GenAI emerged: disuse, misuse, abuse, and the appropriate use. A few participants demonstrated disuse after initially trying, expressing skepticism toward GenAI or choosing not to integrate it into their learning process (e.g., S7 and S29). The second form was misuse, as unintended inappropriate use of GenAI without adequate critical review or revision due to misunderstanding or a lack of expertise (e.g., S1 and S33). Conversely, the third form, abuse, involved the intentional and inappropriate use of GenAI for personal convenience, such as fully relying on GenAI in writing, potentially undermining the originality of the work (e.g., S5). Finally, the appropriate use was identified among students who engaged in iterative feedback with GenAI to refine grammar and style, preserving agency over their final outputs and explicitly acknowledging the use of GenAI in writing (e.g., S37).

Discussion

This study employs an explanatory sequential mixed-methods design grounded in the SOR framework to investigate university students' trust in, reliance on, and resistance to GenAI in language learning. The quantitative analysis performed structural equation modeling to examine the relationships among the stimulus, organism, and response factors within the proposed conceptual model of learners' trust in GenAI. Based on the quantitative findings, the qualitative analysis provided a deeper understanding of the relationships within the refined conceptual model. The findings collectively illustrate university students' trust in, reliance on, and resistance to GenAI for language learning, and boundary conditions that influence their use.

Relationships between university students' trust in GenAI and reliance/resistance in their adoption of GenAI for language learning

Quantitative and qualitative results revealed that trust in GenAI (i.e., intrinsic and extrinsic trust) exerted a positive influence on reliance (i.e., over-reliance, appropriate reliance, and under-reliance), and reliance mediated the relationship between trust in GenAI and behavioral intention to use. This indicates that students with more trust in GenAI tend to rely more on GenAI, and reliance influences the extent to which trust in GenAI affects their behavioral intention to use. The finding aligns with previous research that supports a positive relationship between individuals' trust in AI and reliance (e.g., Choung et al., 2023; Klingbeil et al., 2024). It also echoes previous findings indicating that students who trust AI are more likely to rely on AI recommendations, which in turn increases their willingness to use (Li & Little, 2023). This is probably because individuals' decision-making processes tend to favor fast and optimal solutions over slow and practically constrained solutions (Zhai et al., 2024). Therefore, as AI undertakes roles traditionally played by human cognitive processes, students who trust GenAI advice tend to rely on efficient cognitive shortcuts that facilitate quick and easy decision-making approaches, thereby becoming more likely to use GenAI in language learning (Semrl et al., 2023).

Furthermore, mixed-methods results showed that trust in GenAI exerted a negative effect on resistance (i.e., passive and active resistance), with resistance as a mediator between trust in GenAI and behavioral intention to use. This indicates that students who have more trust in GenAI tend to exhibit less resistance to GenAI, and resistance affects the extent to which trust in GenAI influences behavioral intention to use. This is consistent with previous findings that students who have developed trust and connections with GenAI through interaction are less likely to resist its use and less reluctant to adopt it for learning in online higher education (Aldawsari & Almohish, 2024). Possible reasons for resistance include challenges in interacting with GenAI, concerns about accuracy (Elkefi et al., 2024), fear of the potential negative academic consequences such as plagiarism and undermined thinking skills (Chen & Lin, 2024), and various AI anxieties regarding job replacement and artificial consciousness (Li & Huang, 2020). Thus, university students who have developed trust in GenAI tend to resist using it less, which in turn increases their behavioral intention to use.

The boundary condition of perceived risk for the influence of trust in GenAI on reliance, resistance, and behavioral intention to use

The qualitative results revealed that perceived risk (i.e., academic integrity, data accuracy and security, ethical concerns, and social equity) either mitigates or strengthens the influence of trust in GenAI on reliance, resistance, and behavioral intention to use. This finding indicates that, despite trust, students with more perceived risks associated with using GenAI tend to be more cautious, thereby placing less reliance and behavioral intention to use, and more resistance. Although research on these relationships is scant, existing studies have consistently supported perceived risk as a boundary condition, which explains why trust in GenAI may not result in reliance or behavioral intention to use the technology in practice (e.g., Solberg et al., 2022; Stuck et al., 2020). For people who take risks based on their trust, their trust in AI should exceed their perceived risk in different situations, thereby demonstrating behavioral intention to use (Solberg et al., 2022). The situational risk related to AI involves potential negative impacts on individuals' performance (e.g., failing or losing grades in exams), financial loss, psychological distress (e.g., anxiety or sadness), and various social concerns such as damage to social image, ethical issues, privacy violations, and security threats (Stuck et al., 2020). Specifically, our findings echo the previous research indicating that when students use GenAI in language learning, they have perceived risk on accuracy, transparency, privacy, and ethics, which influence the impact of their trust on GenAI on reliance and behavioral intention to use on the one hand (Chan & Hu, 2023), and on resistance to GenAI on the other (Wang et al., 2024a). It should also be noted that there is a tendency for a risk–benefit calculation, wherein students see risks but are persuaded by the potential benefits of using GenAI (Kreps et al., 2023). In other words, students tend to rely more on GenAI and resist it less not so much because they trust GenAI, but because they perceive that their anticipated benefits of using GenAI will exceed the perceived risk (Ivanov et al., 2024). Other possible reasons include the "fear of missing out," assessments about the lack of efficient alternatives, and optimism about technological advancements (Kreps et al., 2023). Thus, despite the boundary condition of perceived risk, we should remain

cautious about the possibility that students may become over-reliant on GenAI in learning.

The moderating role of self-directed learning and learners' actual use of GenAI

Both quantitative and qualitative results showed a moderating role of self-directed learning (i.e., motivation, self-management, and self-monitoring) between behavioral intention to use and actual use. Specifically, self-directed learning was shown to moderate the effect of behavioral intention to use on actual use. This finding indicates that students with higher levels of self-directed learning are more likely to actualize their intention regarding GenAI use in language learning. This finding aligns with empirical research indicating that effective self-directed learning strategies can improve students' actual utilization of GenAI, making them perceive the use of GenAI as beneficial for enhancing their learning (Wu et al., 2024). GenAI facilitates self-directed learning by designing learning activities, regulating learning plans, and monitoring the learning process (Chiu, 2024). Meanwhile, students with high levels of self-directed learning utilize GenAI in diverse ways, primarily employing it as an interlocutor, a content generator, and a language learning resource (Dizon, 2024). Additionally, it is worth noting that self-directed learning in GenAI-assisted language learning is an autonomous and adaptive process. In the process, learners exercise autonomy to take charge of learning and flexibly adapt interaction with GenAI, by modifying queries, prompts, and modes of engagement in response to their evolving cognitive states (Li et al., 2024). Therefore, self-directed learning either amplifies or diminishes the impact of behavioral intention to use on actual use as an autonomous and adaptive process.

In terms of learners' actual use of GenAI, our quantitative and qualitative results suggested that trust in GenAI influenced actual use. Moreover, four forms of actual use (i.e., disuse, misuse, abuse, and the appropriate use) emerged among the diverse utilities of GenAI in language learning from initial ideation to the refinement of final written outputs. The findings align with prior literature that emphasizes the importance of calibrated trust and the appropriate use of technology (de Visser et al., 2020; Parasuraman & Riley, 1997). Specifically, disuse, reflecting skepticism and avoidance behaviors, supports findings from Trust Calibration Theory, which posits that low trust may prevent beneficial technology use (Lee & See, 2004), thus preventing potential educational gains (Symeou et al., 2025). Conversely, misuse and abuse, representing either unintentional or intentional inappropriate use of GenAI in learning, echo concerns raised in research about the negative consequences associated with over-reliance on GenAI and ethical violations, which may impair learners' cognitive abilities and academic integrity (Abbas et al., 2024; Zhai et al., 2024). However, the appropriate use, characterized by active and iterative interactions with GenAI alongside ethical awareness, critical evaluation, and realization of agency, aligns with previous studies highlighting that calibrated trust in GenAI fosters positive learning experiences or outcomes (de Visser et al., 2020; Symeou et al., 2025). Collectively, these findings highlight that the four forms of actual use are closely related to trust in GenAI, representing a continuum of reliance ranging from high dependence on GenAI to greater learner agency, thereby underscoring the importance of calibrating trust to foster the appropriate use.

Limitations and future research directions

While this study has provided prolific findings, several limitations should be acknowledged to stimulate future research. First, the cross-sectional design restricts the inference of causality among factors, even though the proposed model is grounded in solid theoretical foundations. Future research may consider longitudinal or experimental designs, such as randomized controlled trials, to better clarify the temporal sequences and causal relationships among the factors. This is particularly necessary as students' trust in, reliance on, and resistance to GenAI may change over time as they become more familiar with the technology. Conducting experimental studies to objectively assess the behavioral changes could offer insights into the dynamic changes. Second, the disciplinary, geographic, and cultural homogeneity of our participants, drawn exclusively from the area of language learning within higher education institutions in China, may constrain the generalizability of the findings to other disciplines, educational systems, and cultural contexts. Future research may include participants from diverse disciplinary, geographic, and cultural contexts to enable comparative analyses and explore potential differences. Third, despite our efforts to ensure survey clarity and participant anonymity, self-report surveys have a risk of response bias, potentially influencing the interpretation of the results. For example, students might have underreported their resistance to GenAI due to social desirability bias as a tendency to underreport unfavorable attitudes or behaviors. Future studies could incorporate in-depth interviews or participant observations to triangulate findings obtained through indirect and implicit measures (Bergen & Labonté, 2020). Fourth, this study did not investigate variations in individual student backgrounds or differences across application contexts. Given that students' demographic characteristics (e.g., genders and education levels) and application contexts (e.g., formal and informal settings) may influence their use of GenAI in learning (Strzelecki & ElArabawy, 2024), future research may explore how these differences shape students' trust in, reliance on, and resistance to GenAI in learning.

Implications for theoretical development and educational practices

This study has considerable implications for theoretical development and educational practices. In theory, the study extends the theoretical foundations of trust in, reliance on, and resistance to GenAI by developing and consolidating a refined conceptual model within the SOR framework. It reveals relationships among the factors and enhances understanding of the boundary conditions shaping university students' actual use of GenAI in language learning, thus providing implications for further theoretical development. First, the study expands the SOR framework by incorporating psychological factors (i.e., trust in, reliance on, and resistance to GenAI) and cognitive factors (i.e., self-directed learning and perceived risk) within the context of GenAI-assisted language learning. This expansion improves the applicability of the SOR framework in understanding the impacts of the psychological and cognitive factors on university students' use of GenAI in language learning. Future theoretical models could incorporate environmental factors (e.g., institutional training and policy guidance), technological factors (e.g., task alignment and learner-GenAI interaction), and individual factors (e.g., personality traits and prior experience) to offer a more comprehensive understanding of GenAI

use in education. Second, this study complements Trust Calibration Theory by revealing how university students' trust in GenAI acts as a stimulus that influences reliance and resistance, which subsequently shape their behavioral intention to use and actual use. In this study, trust in GenAI is employed for the first time as an antecedent to explain reliance and resistance in GenAI-assisted language learning. Future theoretical development could explore whether reliance or resistance dynamically influences trust in GenAI as students engage with GenAI over time (Kahr et al., 2024). Furthermore, investigating how varying levels of trust in GenAI might subsequently affect levels of reliance and resistance could further deepen understanding of these dynamic relationships. Third, the study adds to the discourse on self-directed learning by revealing its multi-faceted dimensions and characterizing it as an autonomous and adaptive process in GenAI-assisted language learning. Future theoretical development could examine how GenAI tools facilitate or reshape self-directed learning and investigate potential barriers, such as AI literacy (i.e., the competencies of understanding, applying, critically evaluating, and having awareness of ethical and moral considerations during use of AI) and resource accessibility (Wang et al., 2024a). The investigations will help understand factors influencing whether students' behavioral intention translates into effective actual use.

In educational practices, this study provides practical implications for collaborative efforts among various stakeholders to develop effective strategies for integrating GenAI into higher education. We provide recommendations for conceptualizing trust in GenAI as a complex concept where the goal is to achieve appropriate use of GenAI in education. One key finding of this study is related to the crucial role of trust in GenAI in shaping students' reliance and resistance, thereby influencing their use of GenAI for learning. Fostering appropriate trust in GenAI, which underpins appropriate reliance and ultimately appropriate use, requires collaborative efforts. For one thing, higher education institutions and media sources should cultivate appropriate extrinsic trust by accurately communicating both the learning opportunities GenAI can offer and potential negative issues, such as plagiarism, privacy concerns, and the risk of undermining critical thinking skills (Meryer et al., 2023). For example, higher education institutions should develop and implement clear guidelines regarding the responsible use of GenAI (Moorhouse et al., 2023). Such transparent communication equips students with realistic expectations and informed judgment, enabling them to critically evaluate GenAI outputs and ethically use GenAI in learning. For another, AI developers should aim to foster appropriate intrinsic and extrinsic trust through human-centered GenAI designs that prioritize appropriate trust, instead of greater trust (de Visser et al., 2020). This can be achieved by simplifying GenAI algorithms and operational processes to make user interaction more understandable, and by carefully evaluating ancillary attributes of GenAI to ensure appropriate trust and, consequently, appropriate reliance (Lee & See, 2004). Thus, these efforts could help foster appropriate intrinsic and extrinsic trust, thereby facilitating appropriate reliance.

This study also provides evidence for the moderating effects of perceived risk and self-directed learning on students' actual use, underscoring the importance of raising students' awareness of potential risk and improving their self-directed learning abilities when using GenAI. To raise students' perceived risk, educators should clearly communicate concerns such as privacy issues and plagiarism, guide students to use GenAI as a

supplementary rather than a primary learning resource, and encourage the critical evaluation of information generated by GenAI (Symeou et al., 2025). In addition, to enhance students' self-directed learning abilities, curriculum developers should design customized GenAI-related courses that integrate GenAI into learning activities and guide students in planning, monitoring, and evaluating their learning processes. Additionally, university administrators should establish peer support systems to assist students with limited self-directed learning abilities related to GenAI, enabling them to effectively engage in GenAI-supported learning environments (Esiyok et al., 2024).

This study highlights the importance of addressing disuse, misuse, and abuse of GenAI to ensure the appropriate use of GenAI in educational settings. To this end, first, higher education institutions should develop continuous multi-faceted strategies to stay informed about updates, new GenAI tools, and evolving best practices, thus ensuring the appropriate use of GenAI in education. Given that GenAI is also relatively new to educators, an important strategy includes structured professional training opportunities and experience-sharing sessions in which colleagues with GenAI expertise can share best practices (Symeou et al., 2025). Another strategy is that higher education institutions should develop AI literacy programs to equip students with the knowledge and skills to critically evaluate and effectively utilize GenAI in learning (Yan et al., 2024). As suggested by theories and prior studies, AI literacy is associated with trust in GenAI, and it is essential for promoting the effective, responsible, and ethical use of GenAI in education (e.g., Shahzad et al., 2024; Wang et al., 2024a; Wong et al., 2023). Thus, the training programs may focus on AI literacy, simultaneously familiarizing students with specific tasks and various tools used in different learning contexts (Wong et al., 2023). Second, educators should foster a learning environment that guides students on using GenAI with appropriate reliance, while preventing the potential for over-reliance that may undermine creativity, critical thinking, and problem-solving skills (Dwivedi et al., 2023; Zhai et al., 2024). For example, educators may encourage students to derive satisfaction in mastering challenging concepts independently and find a sense of accomplishment by rewarding students for their genuine intellectual achievements (Abbas et al., 2024). Educators may also promote students' creativity, critical thinking, and problem-solving skills by assigning tasks, activities, or projects that cannot be completed solely by GenAI, and applying diverse criteria to evaluate students' learning (Yan et al., 2024). Third, policymakers should develop clear policies to prevent misuse and abuse in education that ensure appropriate use. Examples of misuse and abuse, potential consequences, and monitoring measures such as plagiarism detection tools should be explicitly communicated (Symeou et al., 2025). Through the collaborative efforts among various stakeholders, higher education institutions may foster the appropriate use of GenAI to better prepare both teachers and students for educational opportunities and challenges in the current era of GenAI.

Conclusion

This study provides a new perspective on understanding university students' trust in, reliance on, and resistance to GenAI in language learning. Based on the SOR framework, this explanatory sequential mixed-methods study has proposed and consolidated a refined conceptual model by exploring the nuanced relationships among the factors

in stimulus (i.e., trust in GenAI), organism (i.e., reliance and resistance), and response (i.e., behavioral intention to use, self-directed learning, and actual use), and boundary conditions among the relationships. The quantitative findings contribute to the development of a proposed conceptual model that statistically explains university students' adoption of GenAI in language learning. Subsequently, the qualitative findings reinforce and refine the conceptual model by revealing forms of trust in GenAI, reliance, resistance, perceived risk, and actual use, and identifying boundary conditions concerning perceived risk and self-directed learning. Results indicate the mediating roles of reliance and resistance in the relationships between trust in GenAI and behavioral intention to use, and the moderating role of self-directed learning in the relationship between behavioral intention to use and actual use. Moreover, forms within the five factors have been revealed, including trust in GenAI (i.e., intrinsic and extrinsic trust), reliance (i.e., over-reliance, appropriate reliance, and under-reliance), resistance (i.e., passive and active resistance), perceived risk (i.e., academic integrity, data accuracy and security, ethical concerns, and social equity), and actual use (i.e., disuse, misuse, abuse, and appropriate use). Additionally, perceived risk is the boundary condition that either mitigates or strengthens the influence of trust in GenAI on reliance, resistance, and behavioral intention to use. Finally, self-directed learning in GenAI-assisted language learning is shown in motivation, self-management, and self-monitoring as an autonomous and adaptive process. These findings shed light on understanding how university students' trust in GenAI interacts with perceived risk and self-directed learning to influence their reliance and resistance, subsequently shaping their actual use of GenAI in education.

Acknowledgements
Not applicable.

Author contributions
Feifei Wang: Conceptualization, data collection, data analysis, and manuscript draft. Nuoen Li: Conceptualization, data collection, data analysis, and manuscript revision. Alan C. K. Cheung: Conceptualization, data analysis, supervision, and manuscript revision. Gary K. W. Wong: Data analysis, interpretation of data, project administration, and manuscript revision.

Funding
Not applicable.

Availability of data and materials
The dataset used in this study is available from the corresponding author upon reasonable request.

Declarations

Competing interests
The authors declare that they have no competing interests.

Received: 31 January 2025 Accepted: 11 June 2025
Published online: 20 August 2025

References

Abbas, M., Jam, F. A., & Khan, T. I. (2024). Is it harmful or helpful? Examining the causes and consequences of generative AI usage among university students. *International Journal of Educational Technology in Higher Education, 21*(10), 1–22. https://doi.org/10.1186/s41239-024-00444-7

Ajzen, I., & Fishbein, M. (1980). *Understanding attitudes and predicting social behavior*. Prentice Hall.

Aldawsari, M. M. M., & Almohish, N. R. I. (2024). Threats and opportunities of students' use of AI-integrated technology (ChatGPT) in online higher education: Saudi Arabian educational technologists' perspectives. *International Review of Research in Open and Distributed Learning, 25*(3), 19–36. https://doi.org/10.19173/irrodl.v25i3.7642

Arce-Urriza, M., Chocarro, R., Cortiñas, M., & Marcos-Matás, G. (2025). From familiarity to acceptance: The impact of generative artificial intelligence on consumer adoption of retail chatbots. *Journal of Retailing and Consumer Services, 84*, Article 104234. https://doi.org/10.1016/j.jretconser.2025.104234

Bai, L. (2025). Research on influencing factors and mechanisms of college students' use of artificial intelligence tools based on SOR and rational behavior models. *Current Psychology, 44*, 1099–1112. https://doi.org/10.1007/s12144-024-07167-y

Balfe, N., Sharples, S., & Wilson, J. R. (2018). Understanding is key: An analysis of factors pertaining to trust in a real-world automation system. *Human Factors, 60*(4), 477–495. https://doi.org/10.1177/0018720818761256

Barber, B. (1983). *The logic and limits of trust*. Rutgers University Press.

Benbasat, I., & Wang, W. (2005). Trust in and adoption of online recommendation agents. *Journal of the Association for Information Systems, 6*(3), 72–101. https://doi.org/10.17705/1jais.00065

Bergen, N., & Labonté, R. (2020). "Everything is perfect, and we have no problems": Detecting and limiting social desirability bias in qualitative research. *Qualitative Health Research, 30*(5), 783–792. https://doi.org/10.1177/1049732319889354

Campbell, S., Greenwood, M., Prior, S., Shearer, T., Walkem, K., Young, S., Bywaters, D., & Walker, K. (2020). Purposive sampling: Complex or simple? Research case examples. *Journal of Research in Nursing, 25*(8), 652–661. https://doi.org/10.1177/1744987120927206

Chan, C. K. Y., & Hu, W. (2023). Students' voices on generative AI: Perceptions, benefits, and challenges in higher education. *International Journal of Educational Technology in Higher Education, 20*(1), 1–18. https://doi.org/10.1186/s41239-023-00411-8

Chavoshi, A., & Hamidi, H. (2019). Social, individual, technological and pedagogical factors influencing mobile learning acceptance in higher education: A case from Iran. *Telematics and Informatics, 38*, 133–165. https://doi.org/10.1016/j.tele.2018.09.007

Chen, J. J., & Lin, J. C. (2024). Artificial intelligence as a double-edged sword: Wielding the POWER principles to maximize its positive effects and minimize its negative effects. *Contemporary Issues in Early Childhood, 25*(1), 146–153. https://doi.org/10.1177/14639491231169813

Cheng, X., Bao, Y., Zarifis, A., Gong, W., & Mou, J. (2022). Exploring consumers' response to text-based chatbots in e-commerce: The moderating role of task complexity and chatbot disclosure. *Internet Research, 32*(2), 496–517. https://doi.org/10.1108/INTR-08-2020-0460

Chiu, T. K. (2024). A classification tool to foster self-regulated learning with generative artificial intelligence by applying self-determination theory: A case of ChatGPT. *Educational Technology Research and Development, 72*(4), 2401–2416. https://doi.org/10.1007/s11423-024-10366-w

Choi, S., Jang, Y., & Kim, H. (2023). Influence of pedagogical beliefs and perceived trust on teachers' acceptance of educational artificial intelligence tools. *International Journal of Human-Computer Interaction, 39*(4), 910–922. https://doi.org/10.1080/10447318.2022.2049145

Chomeya, R. (2010). Quality of psychology test between Likert scale 5 and 6 points. *Journal of Social Sciences, 6*(3), 399–403.

Choung, H., David, P., & Ross, A. (2023). Trust in AI and its role in the acceptance of AI technologies. *International Journal of Human-Computer Interaction, 39*(9), 1727–1739. https://doi.org/10.1080/10447318.2022.2050543

Collier, J. E. (2020). *Applied structural equation modeling using AMOS: Basic to advanced techniques*. Routledge.

Creswell, J. W., & Guetterman, T. C. (2019). *Educational research: Planning, conducting, and evaluating quantitative and qualitative research* (6th ed.). Pearson Education.

Creswell, J. W., & Plano Clark, V. L. (2017). *Designing and conducting mixed methods research* (3rd ed.). Sage.

de Visser, E. J., Peeters, M. M. M., Jung, M. F., Kohn, S., Shaw, T. H., Pak, R., & Neerincx, M. A. (2020). Towards a theory of longitudinal trust calibration in human–robot teams. *International Journal of Social Robotics, 12*(2), 459–478. https://doi.org/10.1007/s12369-019-00596-x

Dergaa, I., Chamari, K., Zmijewski, P., & Saad, H. B. (2023). From human writing to artificial intelligence generated text: Examining the prospects and potential threats of ChatGPT in academic writing. *Biology of Sport, 40*(2), 615–622. https://doi.org/10.5114/biolsport.2023.125623

Dizon, G. (2024). ChatGPT as a tool for self-directed foreign language learning. *Innovation in Language Learning and Teaching*. https://doi.org/10.1080/17501229.2024.2413406

Driscoll, J. W. (1978). Trust and participation in organizational decision making as predictors of satisfaction. *Academy of Management Journal, 21*(1), 44–56. https://doi.org/10.5465/255661

Dwivedi, Y. K., Kshetri, N., Hughes, L., Slade, E. L., Jeyaraj, A., Kar, A. K., Baabdullah, A. M., Koohang, A., Raghavan, V., Ahuja, M., Albanna, H., Albashrawi, M. A., Al-Busaidi, A. S., Balakrishnan, J., Barlette, Y., Basu, S., Bose, I., Brooks, L., Buhalis, D., … Davies, G. H. (2023). "So what if ChatGPT wrote it?" Multidisciplinary perspectives on opportunities, challenges and implications of generative conversational AI for research, practice and policy. *International Journal of Information Management, 71*(102642), 1–63. https://doi.org/10.1016/j.ijinfomgt.2023.102642

Elkefi, S., Tounsi, A., & Kefi, M. A. (2024). Use of ChatGPT for education by engineering students in developing countries: A mixed-methods study. *Behaviour & Information Technology*. https://doi.org/10.1080/0144929X.2024.2354428

Esiyok, E., Gokcearslan, S., & Kucukergin, K. G. (2024). Acceptance of educational use of AI chatbots in the context of self-directed learning with technology and ICT self-efficacy of undergraduate students. *International Journal of Human-Computer Interaction*. https://doi.org/10.1080/10447318.2024.2303557

Fishbein, M., & Ajzen, I. (1975). *Beliefs, attitude, intention, and behavior: An introduction to theory and research*. Addison-Wesley Publishing.

Fraenkel, J. R., Wallen, N. E., & Hyun, H. H. (2018). *How to design and evaluate research in education* (10th ed.). McGraw-Hill Education.

Glikson, E., & Woolley, A. W. (2020). Human trust in artificial intelligence: Review of empirical research. *Academy of Management Annals, 14*(2), 627–660. https://doi.org/10.5465/annals.2018.0057

Gong, J., Said, F., Ting, H., Firdaus, A., Aksar, I. A., & Xu, J. (2023). Do privacy stress and brand trust still matter? Implications on continuous online purchasing intention in China. *Current Psychology, 42*(18), 15515–15527. https://doi.org/10.1007/s12144-022-02857-x

Goyal, S., Chauhan, S., & Gupta, P. (2021). Users' response toward online doctor consultation platforms: SOR approach. *Management Decision, 60*(7), 1990–2018. https://doi.org/10.1108/MD-02-2021-0268

Guo, K., Zhong, Y., Li, D., & Chu, S. K. W. (2023). Effects of chatbot-assisted in-class debates on students' argumentation skills and task motivation. *Computers & Education, 203*(104862), 1–19. https://doi.org/10.1016/j.compedu.2023.104862

Hair, J. F., Black, W. C., Babin, B. J., & Anderson, R. E. (2010). *Multivariate data analysis* (7th ed.). Pearson Prentice Hall.

Higgs, J. M., & Stornaiuolo, A. (2024). Being human in the age of generative AI: Young people's ethical concerns about writing and living with machines. *Reading Research Quarterly, 59*(4), 632–650. https://doi.org/10.1002/rrq.552

Hınız, G. (2024). A year of generative AI in English language teaching and learning - A case study. *Journal of Research on Technology in Education*. https://doi.org/10.1080/15391523.2024.2404132

Israelsen, B. W., & Ahmed, N. R. (2019). "Dave I can assure you that it's going to be all right" A definition, case for, and survey of algorithmic assurances in human-autonomy trust relationships. *ACM Computing Surveys, 51*(6), 1–37. https://doi.org/10.1145/3267338

Ivankova, N. V., Creswell, J. W., & Stick, S. L. (2006). Using mixed-methods sequential explanatory design: From theory to practice. *Field Methods, 18*(1), 3–20. https://doi.org/10.1177/1525822X05282260

Ivanov, S., Soliman, M., Tuomi, A., Alkathiri, N. A., & Al-Alawi, A. N. (2024). Drivers of generative AI adoption in higher education through the lens of the Theory of Planned Behaviour. *Technology in Society, 77*(102521), 1–14. https://doi.org/10.1016/j.techsoc.2024.102521

Jacovi, A., Marasović, A., Miller, T., & Goldberg, Y. (2021). Formalizing trust in artificial intelligence: Prerequisites, causes and goals of human trust in AI. *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, 624–635. https://doi.org/10.1145/3442188.3445923

Johnson, R. B., & Onwuegbuzie, A. J. (2004). Mixed methods research: A research paradigm whose time has come. *Educational Researcher, 33*(7), 14–26. https://doi.org/10.3102/0013189X033007014

Johnson, M., & Vera, A. (2019). No AI is an island: The case for teaming intelligence. *AI Magazine, 40*(1), 16–28. https://doi.org/10.1609/aimag.v40i1.2842

Kahr, P. K., Rooks, G., Willemsen, M. C., & Snijders, C. C. (2024). Understanding trust and reliance development in AI advice: Assessing model accuracy, model explanations, and experiences from previous interactions. *ACM Transactions on Interactive Intelligent Systems, 14*(4), 1–30. https://doi.org/10.1145/3686164

Kelly, S., Kaye, S. A., & Oviedo-Trespalacios, O. (2023). What factors contribute to the acceptance of artificial intelligence? *A Systematic Review. Telematics and Informatics, 77*, Article 101925. https://doi.org/10.1016/j.tele.2022.101925

Kendall, G., & Teixeira da Silva, J. A. (2023). Risks of abuse of large language models, like ChatGPT, in scientific publishing: Authorship, predatory publishing, and paper mills. *Learned Publishing*. https://doi.org/10.1002/leap.1578

Kim, H. W., & Kankanhalli, A. (2009). Investigating user resistance to information systems implementation: A status quo bias perspective. *MIS Quarterly, 33*(3), 567–582. https://doi.org/10.2307/20650309

Kline, R. B. (2023). *Principles and practice of structural equation modeling* (5th ed.). Guilford.

Klingbeil, A., Grützner, C., & Schreck, P. (2024). Trust and reliance on AI—An experimental study on the extent and costs of overreliance on AI. *Computers in Human Behavior, 160*(108352), 1–10. https://doi.org/10.1016/j.chb.2024.108352

Komiak, S. Y. X., & Benbasat, I. (2006). The effects of personalization and familiarity on trust and adoption of recommendation agents. *MIS Quarterly, 30*(4), 941–960. https://doi.org/10.2307/25148760

Kreps, S., George, J., Lushenko, P., & Rao, A. (2023). Exploring the artificial intelligence "trust paradox": Evidence from a survey experiment in the United States. *PLoS ONE, 18*(7), Article e0288109. https://doi.org/10.1371/journal.pone.0288109

Küper, A., & Krämer, N. (2024). Psychological traits and appropriate reliance: Factors shaping trust in AI. *International Journal of Human-Computer Interaction*. https://doi.org/10.1080/10447318.2024.2348216

Lai, Y., Saab, N., & Admiraal, W. (2022). University students' use of mobile technology in self-directed language learning: Using the integrative model of behavior prediction. *Computers & Education, 179*(104413), 1–13. https://doi.org/10.1016/j.compedu.2021.104413

Law, L. (2024). Application of generative artificial intelligence (GenAI) in language teaching and learning: A scoping literature review. *Computers and Education Open, 6*(100174), 1–13. https://doi.org/10.1016/j.caeo.2024.100174

Lee, J. D., & See, K. A. (2004). Trust in automation: Designing for appropriate reliance. *Human Factors, 46*(1), 50–80. https://doi.org/10.1518/hfes.46.1.50_30392

Li, J., & Huang, J. S. (2020). Dimensions of artificial intelligence anxiety based on the integrated fear acquisition theory. *Technology in Society, 63*(101410), 1–10. https://doi.org/10.1016/j.techsoc.2020.101410

Li, M. D., & Little, B. P. (2023). Appropriate reliance on artificial intelligence in radiology education. *Journal of the American College of Radiology, 20*(11), 1126–1130. https://doi.org/10.1016/j.jacr.2023.04.019

Li, B., Bonk, C. J., Wang, C., & Kou, X. (2024). Reconceptualizing self-directed learning in the era of generative AI: An exploratory analysis of language learning. *IEEE Transactions on Learning Technologies, 17*, 1515–1529. https://doi.org/10.1109/TLT.2024.3386098

Liu, G., & Ma, C. (2023). Measuring EFL learners' use of ChatGPT in informal digital learning of English based on the technology acceptance model. *Innovation in Language Learning and Teaching, 18*(2), 125–138. https://doi.org/10.1080/17501229.2023.2240316

Liu, I. F., Chen, R. S., & Lu, H. C. (2015). An exploration into improving examinees' acceptance of participation in an online exam. *Educational Technology & Society, 18*(2), 153–165.

Liu, G. L., Darvin, R., & Ma, C. (2024). Exploring AI-mediated informal digital learning of English (AI-IDLE): A mixed-method investigation of Chinese EFL learners' AI adoption and experiences. *Computer Assisted Language Learning*. https://doi.org/10.1080/09588221.2024.2310288

Lorenz, P., Perset, K., & Berryhill, J. (2023). Initial policy considerations for generative artificial intelligence. *OECD Artificial Intelligence Papers, 1*, 1–40. https://doi.org/10.1787/fae2d1e6-en

Lounsbury, J. W., Levy, J. J., Park, S. H., Gibson, L. W., & Smith, R. (2009). An investigation of the construct validity of the personality trait of self-directed learning. *Learning and Individual Differences, 19*(4), 411–418. https://doi.org/10.1016/j.lindif.2009.03.001

Ma, M. (2024). Exploring the acceptance of generative artificial intelligence for language learning among EFL postgraduate students: An extended TAM approach. *International Journal of Applied Linguistics*. https://doi.org/10.1111/ijal.12603

Mahmud, H., Islam, A. N., Ahmed, S. I., & Smolander, K. (2022). What influences algorithmic decision-making? A systematic literature review on algorithm aversion. *Technological Forecasting and Social Change, 175*(121390), 1–26. https://doi.org/10.1016/j.techfore.2021.121390

Marsh, H. W., Wen, Z., & Hau, K. T. (2004). Structural equation models of latent interactions: Evaluation of alternative estimation strategies and indicator construction. *Psychological Methods, 9*(3), 275–300. https://doi.org/10.1037/1082-989X.9.3.275

Mayer, R. C., Davis, J. H., & Schoorman, F. D. (1995). An integrative model of organizational trust. *Academy of Management Review, 20*(3), 709–734. https://doi.org/10.5465/amr.1995.9508080335

Mcknight, D. H., Carter, M., Thatcher, J. B., & Clay, P. F. (2011). Trust in a specific technology: An investigation of its components and measures. *ACM Transactions on Management Information Systems (TMIS), 2*(2), 1–25. https://doi.org/10.1145/1985347.1985353

Mehrabian, A., & Russell, J. A. (1974). *An approach to environmental psychology*. MIT Press.

Merritt, S. M., Heimbaugh, H., LaChapell, J., & Lee, D. (2013). I trust it, but I don't know why: Effects of implicit attitudes toward automation on trust in an automated system. *Human Factors, 55*(3), 520–534. https://doi.org/10.1177/0018720812465081

Meyer, J. G., Urbanowicz, R. J., Martin, P. C. N., O'Connor, K., Li, R., Peng, P. C., Bright, T. J., Tatonetti, N., Won, K. J., Gonzalez-Hernandez, G., & Moore, J. H. (2023). ChatGPT and large language models in academia: Opportunities and challenges. *BioData Mining, 16*(20), 1–11. https://doi.org/10.1186/s13040-023-00339-9

Moorhouse, B. L., Yeo, M. A., & Wan, Y. (2023). Generative AI tools and assessment: Guidelines of the world's top-ranking universities. *Computers and Education Open, 5*(15), Article 100151. https://doi.org/10.1016/j.caeo.2023.100151

Naeem, M., Ozuem, W., Howell, K., & Ranfagni, S. (2023). A step-by-step process of thematic analysis to develop a conceptual model in qualitative research. *International Journal of Qualitative Methods, 22*, 1–18. https://doi.org/10.1177/16094069231205789

Nazaretsky, T., Ariely, M., Cukurova, M., & Alexandron, G. (2022). Teachers' trust in AI-powered educational technology and a professional development program to improve it. *British Journal of Educational Technology, 53*(4), 914–931. https://doi.org/10.1111/bjet.13232

Orbaiz, M. L. V., & Arce-Urriza, M. (2024). The role of active and passive resistance in new technology adoption by final consumers: The case of 3D printing. *Technology in Society, 77*(102500), 1–22. https://doi.org/10.1016/j.techsoc.2024.102500

Pack, A., & Maloney, J. (2023). Using generative artificial intelligence for language education research: Insights from using OpenAI's ChatGPT. *TESOL Quarterly, 57*(4), 1571–1582. https://doi.org/10.1002/tesq.3253

Parasuraman, R., & Riley, V. (1997). Humans and automation: Use, misuse, disuse, abuse. *Human Factors, 39*(2), 230–253. https://doi.org/10.1518/001872097778543886

Poppo, L., Zhou, K. Z., & Li, J. J. (2016). When can you trust "trust"? Calculative trust, relational trust, and supplier performance. *Strategic Management Journal, 37*(4), 724–741. https://doi.org/10.1002/smj.2374

Roberts, K., Dowell, A., & Nie, J. B. (2019). Attempting rigour and replicability in thematic analysis of qualitative research data: A case study of codebook development. *BMC Medical Research Methodology, 19*(66), 1–8. https://doi.org/10.1186/s12874-019-0707-y

Santiago, C. S., Jr., Embang, S. I., Conlu, M. T. N., Acanto, R. B., Lausa, S. M., Ambojia, K. W. P., Laput, E. Y., Aperocho, M. D. B., Malabag, B. A., Balilo, B. B., Jr., Paderes, J. J., Cahapin, E. L., & Romasanta, J. K. N. (2023). Utilization of writing assistance tools in research in selected higher learning institutions in the Philippines: A text mining analysis. *International Journal of Learning, Teaching and Educational Research, 22*(11), 259–284. https://doi.org/10.26803/ijlter.22.11.14

Scantamburlo, T., Cortés, A., Foffano, F., Barrué, C., Distefano, V., Pham, L., & Fabris, A. (2025). Artificial intelligence across Europe: A study on awareness, attitude and trust. *IEEE Transactions on Artificial Intelligence, 6*(2), 477–490. https://doi.org/10.1109/TAI.2024.3461633

Schelble, B. G., Lopez, J., Textor, C., Zhang, R., McNeese, N. J., Pak, R., & Freeman, G. (2024). Towards ethical AI: Empirically investigating dimensions of AI ethics, trust repair, and performance in human-AI teaming. *Human Factors, 66*(4), 1037–1055. https://doi.org/10.1177/00187208221116952

Schemmer, M., Kuehl, N., Benz, C., Bartos, A., & Satzger, G. (2023). Appropriate reliance on AI advice: Conceptualization and the effect of explanations. *Proceedings of the 28th International Conference on Intelligent User Interfaces*. https://doi.org/10.1145/3581641.3584066

Schreiber, J. B., Nora, A., Stage, F. K., Barlow, E. A., & King, J. (2006). Reporting structural equation modeling and confirmatory factor analysis results: A review. *The Journal of Educational Research, 99*(6), 323–338. https://doi.org/10.3200/JOER.99.6.323-338

Semrl, N., Feigl, S., Taumberger, N., Bracic, T., Fluhr, H., Blockeel, C., & Kollmann, M. (2023). AI language models in human reproduction research: Exploring ChatGPT's potential to assist academic writing. *Human Reproduction, 38*(12), 2281–2288. https://doi.org/10.1093/humrep/dead207

Seo, K., Tang, J., Roll, I., Fels, S., & Yoon, D. (2021). The impact of artificial intelligence on learner-instructor interaction in online learning. *International Journal of Educational Technology in Higher Education, 18*(54), 1–23. https://doi.org/10.1186/s41239-021-00292-9

Shahzad, M. F., Xu, S., & Zahid, H. (2024). Exploring the impact of generative AI-based technologies on learning performance through self-efficacy, fairness & ethics, creativity, and trust in higher education. *Education and Information Technologies*. https://doi.org/10.1007/s10639-024-12949-9

Shapiro, S. P. (1987). The social control of impersonal trust. *American Journal of Sociology, 93*(3), 623–658. https://doi.org/10.1086/228791

Solberg, E., Kaarstad, M., Eitrheim, M. H. R., Bisio, R., Reegård, K., & Bloch, M. (2022). A conceptual model of trust, perceived risk, and reliance on AI decision aids. *Group & Organization Management, 47*(2), 187–222. https://doi.org/10.1177/10596011221081238

Song, L., & Hill, J. R. (2007). A conceptual model for understanding self-directed learning in online environments. *Journal of Interactive Online Learning, 6*(1), 27–42.

Strzelecki, A. (2023). To use or not to use ChatGPT in higher education? A study of students' acceptance and use of technology. *Interactive Learning Environments, 32*(9), 5142–5155. https://doi.org/10.1080/10494820.2023.2209881

Strzelecki, A., & ElArabawy, S. (2024). Investigation of the moderation effect of gender and study level on the acceptance and use of generative AI by higher education students: Comparative evidence from Poland and Egypt. *British Journal of Educational Technology, 55*(3), 1209–1230. https://doi.org/10.1111/bjet.13425

Stuck, R. E., Holthausen, B. E., & Walker, B. N. (2020). The role of risk in human-robot trust. In C. S. Nam & J. B. Lyons (Eds.), *Trust in human-robot interaction* (pp. 179–194). Elsevier.

Sun, P. P., & Mei, B. (2022). Modeling preservice Chinese-as-a-second/foreign-language teachers' adoption of educational technology: A technology acceptance perspective. *Computer Assisted Language Learning, 35*(4), 816–839. https://doi.org/10.1080/09588221.2020.1750430

Symeou, L., Louca, L., Kavadella, A., Mackay, J., Danidou, Y., & Raffay, V. (2025). Development of evidence-based guidelines for the integration of generative AI in university education through a multidisciplinary, consensus-based approach. *European Journal of Dental Education, 29*(2), 285–303. https://doi.org/10.1111/eje.13069

Vafaei-Zadeh, A., Nikbin, D., Wong, S. L., & Hanifah, H. (2024). Investigating factors influencing AI customer service adoption: An integrated model of stimulus–organism–response (SOR) and task-technology fit (TTF) theory. *Asia Pacific Journal of Marketing and Logistics*. https://doi.org/10.1108/APJML-05-2024-0570

van Dongen, K., & van Maanen, P. P. (2013). A framework for explaining reliance on decision aids. *International Journal of Human-Computer Studies, 71*(4), 410–424. https://doi.org/10.1016/j.ijhcs.2012.10.018

Venkatesh, V., Thong, J. Y. L., & Xu, X. (2012). Consumer acceptance and use of information technology: Extending the unified theory of acceptance and use of technology. *MIS Quarterly, 36*(1), 157–178. https://doi.org/10.2307/41410412

Venkatesh, V., Brown, S. A., & Sullivan, Y. W. (2016). Guidelines for conducting mixed-methods research: An extension and illustration. *Journal of the Association for Information Systems, 17*(7), 435–494. https://doi.org/10.17705/1jais.00433

Wang, C., Wang, H., Li, Y., Dai, J., Gu, X., & Yu, T. (2024a). Factors influencing university students' behavioral intention to use generative artificial intelligence: Integrating the theory of planned behavior and AI literacy. *International Journal of Human-Computer Interaction*. https://doi.org/10.1080/10447318.2024.2383033

Wang, F., Cheung, A. C., & Chai, C. S. (2024b). Language learning development in human-AI interaction: A thematic review of the research landscape. *System, 125*(103424), 1–24. https://doi.org/10.1016/j.system.2024.103424

Wang, F., Cheung, A. C., Chai, C. S., & Liu, J. (2024c). Development and validation of the perceived interactivity of learner-AI interaction scale. *Education and Information Technologies*. https://doi.org/10.1007/s10639-024-12963-x

Wong, G. K., Reichert, F., & Law, N. (2023). Reorienting the assessment of digital literacy in the twenty-first century: A product-lifecycle and experience dependence perspective. *Educational Technology Research and Development, 71*(6), 2389–2412. https://doi.org/10.1007/s11423-023-10278-1

Wu, D., Zhang, S., Ma, Z., Yue, X. G., & Dong, R. K. (2024). Unlocking potential: Key factors shaping undergraduate self-directed learning in AI-enhanced educational environments. *Systems, 12*(9), 1–19. https://doi.org/10.3390/systems12090332

Xu, W., & Zammit, K. (2020). Applying thematic analysis to education: A hybrid approach to interpreting data in practitioner research. *International Journal of Qualitative Methods, 19*, 1–9. https://doi.org/10.1177/1609406920918810

Yan, L., Greiff, S., Teuber, Z., & Gašević, D. (2024). Promises and challenges of generative artificial intelligence for human learning. *Nature Human Behaviour, 8*(10), 1839–1850. https://doi.org/10.1038/s41562-024-02004-5

Zhai, C., Wibowo, S., & Li, L. D. (2024). The effects of over-reliance on AI dialogue systems on students' cognitive abilities: A systematic review. *Smart Learning Environments, 11*(1), 1–37. https://doi.org/10.1186/s40561-024-00316-7

Zhang, C., Rice, R. E., & Wang, L. H. (2024a). College students' literacy, ChatGPT activities, educational outcomes, and trust from a digital divide perspective. *New Media & Society*. https://doi.org/10.1177/14614448241301741

Zhang, R., Zou, D., & Cheng, G. (2024b). Chatbot-based learning of logical fallacies in EFL writing: Perceived effectiveness in improving target knowledge and learner motivation. *Interactive Learning Environments, 32*(9), 5552–5569. https://doi.org/10.1080/10494820.2023.2220374

Zhao, Y., Wang, A., & Sun, Y. (2020). Technological environment, virtual experience, and MOOC continuance: A stimulus–organism–response perspective. *Computers & Education, 144*(103721), 1–14. https://doi.org/10.1016/j.compedu.2019.103721

Zheng, Y., Wang, Y., Liu, K. S. X., & Jiang, M. Y. C. (2024). Examining the moderating effect of motivation on technology acceptance of generative AI for English as a foreign language learning. *Education and Information Technologies, 29*(17), 23547–23575. https://doi.org/10.1007/s10639-024-12763-3

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.